

BACHELOR'S THESIS

Measuring Borrower-Side Concessionality: Constructing a Sovereign Market-Benchmark Dataset, 2012–2024

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Abstract

This thesis examines how official-loan concessionality changes when repayments are assessed against borrowers' market financing conditions rather than the OECD's standardized discount rates of 9%, 7% or 6%, which depend on recipient category. It constructs a sovereign borrowing-rate dataset for 2012–2024, combining bond yields, reported borrowing terms and estimates based on credit ratings or comparable countries. The dataset supplies a reference for 1,701 of 1,756 low- and middle-income country-years, with estimates accounting for much of the coverage. In matched 2018–2024 samples, rates from outstanding bonds, reported borrowing terms, credit ratings and comparable countries each approximate observed issuance more closely, on average, than policy rates. Performance in measuring annual changes is weaker. The loan analysis uses reconstructed repayments and annual rates under a common-currency assumption, holding schedules fixed to compare grant elements, defined as the difference between the advance and the present value of repayments as a percentage of the advance. Across 1,274 usable OECD Creditor Reporting System financing records from 2018–2024, excluding records whose benchmark relies solely on comparable countries, the mean grant element falls from 51.73% under policy rates to 46.09% under borrower-specific rates. Borrower-specific rates nevertheless produce higher grant elements in 36.7% of records. Additional creditor comparisons show that repayment timing can reverse the ordering suggested by contractual interest rates. The findings establish the importance of borrower financing conditions for measuring concessionality and provide a basis for further research on development-finance effectiveness.

Abbreviations

ADD — African Debt Database

CRS — Creditor Reporting System

DAC — Development Assistance Committee

FRED — Federal Reserve Economic Data

GDP — Gross domestic product

GNI — Gross national income

IBRD — International Bank for Reconstruction and Development

IDA — International Development Association

IDS — International Debt Statistics

IMF — International Monetary Fund

LDC — Least developed country

LIC — Low-income country

LMIC — Lower-middle-income country in DAC rate classifications

LSEG — London Stock Exchange Group

MAE — Mean absolute error

NCBP — Non-Concessional Borrowing Policy

ODA — Official development assistance

OECD — Organisation for Economic Co-operation and Development

PVR — Present-value repayment ratio

RMSE — Root mean squared error

SDFP — Sustainable Development Finance Policy

UMIC — Upper-middle-income country

USD — United States dollar

WGI — Worldwide Governance Indicators

1 Introduction

Official development loans differ in the financial advantage they offer recipients. A loan’s amount describes the resources committed, but its interest payments and repayment dates determine the obligations attached to those resources. Concessionality measures how favourable those terms are relative to a specified financing alternative. The grant element expresses the difference between the amount advanced and the present value of repayments as a percentage of the advance. Its value therefore depends on the discount rate used to assess those repayments (Scott 2017; Potts and Chung 2008).

Official aid statistics use standardized discount rates to calculate grant elements. Under the OECD Development Assistance Committee’s grant-equivalent rules for sovereign loans, these rates are 9%, 7% or 6%, depending on the recipient category (OECD n.d.-c). These rules became the headline basis for official development assistance reporting for 2018 flows, which is why the main institutional comparison in this thesis covers 2018–2024 (OECD 2019). Countries within a recipient category can nevertheless face very different market borrowing costs (OECD 2025, 118–122). From the recipient’s perspective, the relevant comparison concerns how the official loan terms compare with its own financing alternatives. Establishing that comparison would give assessments of development lending a more specific financial basis, including public discussion of the terms offered by different creditors.

This thesis asks two linked questions: how can transparent, borrower-specific sovereign borrowing benchmarks be constructed for 2012–2024, and how does using them change measured concessionality relative to policy reference rates? Here, a borrowing benchmark is an observed or estimated annual interest-rate reference intended to approximate a government’s borrowing conditions. The loan comparisons value reconstructed repayments under these references; they do not establish that equivalent market financing was available at each loan’s currency, maturity and commitment date. Constructing these rates is a substantial part of the research because many countries do not issue international bonds every year, and available borrowing information differs in its timing, instrument coverage and completeness. Studies of market participation show that observed issuers are a selected group of developing countries (Gelos, Sahay and Sandleris 2011; Presbitero et al. 2016).

The approach builds on the OECD’s analysis of sovereign bond markets and ONE’s comparison of official repayments with borrower financing conditions (OECD 2025; Harcourt, Haro Ruiz and Rivera 2026; ONE Data n.d.). This thesis constructs an annual dataset combining yields on newly issued and outstanding bonds, World Bank reports of interest rates on new external borrowing, and estimates based on sovereign credit ratings or comparable countries. It preserves each source separately and evaluates the alternative rates against observed issuance on matching countries and years. It then uses the dataset to examine borrowing conditions across countries and over time, and to recalculate grant elements for official loans. The OECD Creditor Reporting System (CRS) supplies the broadest loan sample, supplemented by the records used by Morris, Parks and Gardner (2020), later AidData records and the African Debt Database (AidData 2025; African Debt Database 2026). Within each loan comparison, the repayment schedule remains fixed and only the discount rate changes.

The constructed dataset provides a borrowing reference for 1,701 of 1,756 low- and middle-income country-years. Much of this coverage depends on estimates, particularly for low-income countries. In matched comparisons for 2018–2024, rates from outstanding bonds, reported new external borrowing, credit ratings and comparable countries each approximate yields on newly issued bonds more closely, on average, than the differentiated policy rates. Evidence is stronger for approximating annual rate levels than for measuring changes between years. The country comparisons show a widespread increase in borrowing costs in 2022, accompanied by wider differences between lower- and higher-cost borrowers. These findings support using borrower-specific information while showing why the source and composition of each comparison need to remain visible.

For 1,274 usable CRS financing records from 2018–2024, excluding records whose borrowing benchmark is based solely on comparable countries, the mean grant element falls from 51.73% under the differentiated policy rates to 46.09% under borrower-specific rates. The difference of 5.64 percentage points conceals substantial variation: borrower-specific rates produce a higher grant element in 36.7% of records. The later-year results also differ across creditor portfolios, and the timing of repayments reverses some creditor orderings based on contractual interest rates alone. These results provide a basis for examining concessionality in research on development effectiveness, without treating the financial concession itself as evidence of additional development outcomes.

Chapter 2 develops the concepts, institutional uses and related research. Chapter 3 explains construction of the borrowing-rate dataset, Chapter 4 evaluates its coverage and accuracy, and Chapter 5 examines borrowing conditions across countries and years. Chapter 6 sets out the official-loan data and repayment calculations, followed by the concessionality findings in Chapter 7. Chapter 8 discusses their implications, limitations and contribution to understanding development finance.

2 Concepts, institutional uses and related research

2.1 Concessionality in development finance

Development finance encompasses financial resources used to support economic and social development, including public funds, private investment, grants and remittances (OECD n.d.-a). Official lending, the focus of this thesis, forms part of that wider field. In its economic sense, concessionality describes the degree to which financing is offered on more favourable terms than comparable market financing. For a loan, this assessment considers the amounts and timing of all scheduled payments, including principal and interest. Grants create no repayment obligation, while concessional loans are repayable on preferential terms (OECD n.d.-b, “Concessionality level” and “Grant element”).

A loan’s face value measures how much financing is provided, but it does not capture the concession embedded in its repayment terms. Cordella and Ulku (2004, 3–5) formalize this distinction by holding the donor’s net financial cost constant. In their framework, a donor can provide a larger loan on less favourable terms because a larger share is repaid in present-value terms, or a smaller loan on more favourable terms.

Grant-element measurement developed in the 1960s partly to address this distinction between financing volumes and donors’ net financial contributions. As Scott (2017, 13–14) recounts, Pincus’s 1963 proposal deducted the value of expected repayments when measuring donor effort. Ohlin’s subsequent work developed the method alongside concerns about the indebtedness created by development lending. The historical motivation thus involved both comparing donor contributions and understanding the significance of repayment terms for recipients.

Concessionality measures subsequently became part of development-finance policy and accounting. Within the OECD Development Assistance Committee’s statistics, grant elements help determine which loans qualify as official development assistance (ODA). Under the grant-equivalent reporting system introduced for 2018 flows, they also help determine how much lending counts as donor effort (OECD n.d.-c). These conventions affect comparisons of donor contributions and the incentives attached to different lending terms. Roodman (2015, 399, 403–406) examines how aid-accounting rules can reward some lending terms over others, connecting measurement choices to the incentives facing lenders.

For recipients, the amount of financing received, the amount recorded as aid and the obligations to repay describe different aspects of a lending relationship. This distinction also matters for public discussion of sovereign lending. Assessing the repayment obligations relative to the borrower’s financing conditions provides a financial basis for discussing how favourable the terms are from the recipient’s perspective (Leipziger 1983, 334–335; Potts and Chung 2008, 1027–1028).

2.2 Measuring concessionality from lender and borrower perspectives

A present-value comparison brings the amounts and timing of a loan’s payments into a common measure. For a loan of L fully advanced at the outset, let C_t denote the scheduled payment t years after the advance, including principal, interest and any included fees. The annual discount rate r is expressed as a decimal. Following ONE Data’s *Priced Out* methodology, this thesis reports the present value of repayments per 100 lent as the present-value repayment ratio (PVR). ONE applies a borrower-market discount rate to this measure (ONE Data n.d., “The Present Value Repayment Ratio”; Harcourt, Haro Ruiz and Rivera 2026). The repayment value, PVR and grant element can be written together as

$$V(r) = \sum_t \frac{C_t}{(1+r)^t}, \quad \text{PVR}_{100}(r) = 100 \frac{V(r)}{L}, \quad \text{GE}_{\%}(r) = 100 \left(1 - \frac{V(r)}{L} \right). \quad (1)$$

The grant-equivalent amount is $L - V(r)$. PVR expresses the present value of repayments per 100 advanced, while the grant element expresses the corresponding concession as a percentage. The two measures express the same valuation and depend on the same repayment schedule and discount rate (Scott 2017, sections 1.2–1.4; ONE Data n.d., “The Present Value Repayment Ratio”). This thesis calculates these values using both borrower-market rates and policy reference rates.

The contractual interest rate determines interest owed, whereas the discount rate determines how future payments are valued today. The relevant financing alternative can differ between lender and borrower, so the donor’s financial cost and the recipient’s financing advantage need not coincide. Potts and Chung

(2008, 1027–1029, 1031–1034) develop the recipient perspective further by examining exchange rates, purchasing power and tied procurement, which restricts where the financed goods and services can be purchased. These features affect the value of the resources received as well as the financial terms of the loan.

An earlier implementation of donor-cost valuation is Chang, Fernández-Arias and Servén’s (1999, 3–8, 11–13) *Effective Development Assistance* measure, which combines grants with the grant equivalents of loans to assess donors’ voluntary net financial cost. Their discount rates represent what the lender could earn by investing the same funds elsewhere. They use returns on safe government securities, differentiated by currency, year and payment horizon, without adding a premium for the borrowing country’s credit risk. The resulting grant equivalents measure the donor’s financial concession relative to those alternative investment returns.

This thesis uses borrower-market rates because its empirical question concerns how official repayment obligations compare with financing conditions associated with the borrowing country. The country-specific financing premia omitted from EDA’s safe investment returns are part of that comparison. The currency and maturity detail of EDA’s curves nevertheless remains relevant when assessing how closely a rate matches the loan being valued. Currency, timing, payment horizon and evidence about market access all affect whether an observed or estimated market rate represents a comparable financing alternative (Chang, Fernández-Arias and Servén 1999, 11–13; Potts and Chung 2008, 1027–1029, 1034 n. 6).

Using a common policy discount rate or borrower-specific market rates leads to different comparisons when two governments receive loans on identical terms but face different commercial borrowing costs. Valuing both loans at the same policy rate assigns them the same grant element. Borrower-specific rates instead show how those terms compare with financing conditions in each country. Applying both types of discount rate to the same repayment schedule isolates the valuation difference attributable to the choice of rate (Harcourt, Haro Ruiz and Rivera 2026).

For unchanged official loan terms, a fixed policy discount rate produces an unchanged grant element, whereas changes in the borrower-market rate alter the market-relative grant element. If a borrower’s market rate rises while official loan terms stay fixed, the repayments receive a lower present value and the market-relative grant element increases. The official loan compares more favourably because market financing has become more expensive. A rise in measured concessionality can therefore reflect worsening market borrowing conditions even when the official lender has not changed its terms (ONE Data n.d., note 4). Changes in the estimated concession must consequently be read alongside changes in both official lending terms and market borrowing conditions.

2.3 Concessionality in official reporting and lending policy

Before 2018, the OECD Development Assistance Committee (DAC) used the grant element to determine whether a loan qualified as official development assistance (ODA) within a system based on financial flows. A loan needed a grant element of at least 25%, calculated using a 10% discount rate, as well as meeting the other ODA conditions. Once a loan qualified, its disbursements were counted in full, and

subsequent principal repayments were deducted from net ODA. The grant element therefore determined eligibility but did not determine the fraction of each disbursement counted as aid (OECD DAC 2014, Annex 2, paragraphs 3–4; Scott 2017, section 1.7). The remainder of this section examines the subsequent reform of aid accounting and the use of concessionality in setting borrowing limits and financing terms at the International Monetary Fund and the World Bank.

In December 2014, the DAC agreed to change both the discount rates used to assess loan concessionality and the way qualifying loans were counted. The reform aimed to improve the comparability of donor effort and encourage more favourable lending terms for poorer countries. The decision provided for reporting under both the cash-flow and grant-equivalent systems during 2015–2017 (OECD DAC 2014, Annex 2, paragraphs 5–10). In April 2019, the OECD published its first headline aid figures on a grant-equivalent basis, covering flows in 2018 (OECD 2019).

Under the differentiated rules, bilateral loans to the official sector in least developed countries and other low-income countries are assessed using a 9% discount rate and must have a grant element of at least 45% to qualify as ODA. For lower-middle-income countries, the corresponding rate and minimum grant element are 7% and 15%, and for upper-middle-income countries they are 6% and 10%. The discount rate is used to calculate the present value of scheduled repayments, from which the grant element is obtained. That grant element must meet the recipient group’s minimum. The financing must also be provided by official agencies, go to a recipient on the DAC list and have the promotion of economic development and welfare as its main objective (OECD n.d.-c, “ODA flows”).

Under grant-equivalent reporting, the amount recorded as ODA is calculated by multiplying each year’s disbursements by the loan’s grant element, assessed when the loan was committed. The grant element thus affects both eligibility and the amount of lending recorded as donor effort (OECD n.d.-c, “ODA grant equivalent measure”). The 9%, 7% and 6% discount rates combine a common 5% base with additions of four, two and one percentage points, respectively. In adopting these differentiated rates, the DAC sought to reflect both the funding cost and the risk involved in lending (OECD n.d.-c, note 2; OECD DAC 2014, Annex 2, paragraphs 6–7).

In programmes supported by the International Monetary Fund (IMF), concessionality can affect the amount of new borrowing a government is allowed to undertake or guarantee. The IMF and World Bank adopted a unified 5% discount rate for external debt analysis and loan concessionality in low-income countries in October 2013 (IMF 2013). Under the IMF’s Debt Limits Policy, a loan is generally classified as concessional if its grant element is at least 35% (IMF 2021, 53).

A ceiling on nonconcessional borrowing counts the face value of each covered loan whose grant element is below the programme’s applicable minimum threshold, which is usually 35% but can be set higher. Loans meeting the applicable threshold do not count against that particular limit. A present-value ceiling instead covers both concessional and nonconcessional borrowing (IMF 2021, 30–31, 43, 53). Each loan counts at the present value of its scheduled repayments if that value is below the amount borrowed, and at face value otherwise. More favourable repayment terms can therefore allow a government to obtain

more financing within a given present-value ceiling (IMF 2021, 59). The programme specifies which borrowing is covered and any exemptions (IMF 2021, 38, 40, 42–43).¹

A country’s debt position also influences the financing it receives from the World Bank’s International Development Association (IDA). Among eligible borrowers, assessed debt-distress risk helps determine whether IDA provides grants or concessional loans and the terms of those loans. These decisions affect whether the financing must be repaid and, where repayment is required, the obligations the country assumes (World Bank 2024).

IDA also sets conditions on borrowing from other creditors. Its Sustainable Development Finance Policy (SDFP) replaced the Non-Concessional Borrowing Policy (NCBP) in July 2020 (World Bank 2020, 20–22; Chuhan-Pole 2021). Agreed policy actions can include ceilings on nonconcessional external borrowing. Under the SDFP, unsatisfactory implementation can result in part of a country’s allocation being set aside; if performance lags for two years, the first year’s set-aside is lost. In some cases, severe or repeated breaches of policy actions, such as a nonconcessional borrowing ceiling, can also lead to harder IDA financing terms (World Bank 2020, 9, 14). The DAC requires loans reported as ODA to comply with the applicable IMF and World Bank borrowing policies (OECD n.d.-c, “ODA flows”).

Table 1: Concessionality in aid reporting and borrowing policy

Institutional use	Convention and period	Consequence
DAC loan eligibility under the former headline system	10% discount rate and minimum 25% grant element. Headline system through 2017.	Qualifying disbursements count in full as ODA. Principal repayments reduce net ODA.
DAC bilateral official-sector loans under grant-equivalent reporting	Discount rate / minimum grant element: LDCs and other LICs, 9% / 45%. LMICs, 7% / 15%. UMICs, 6% / 10%. Transition in 2015–2017, headline basis from 2018.	The grant element determines eligibility and the share of disbursements counted as donor effort.
IMF programme debt limits	Unified 5% discount rate from October 2013. Minimum grant-element threshold usually 35%, but a programme may set it higher.	Ceilings limit covered new borrowing through its face value or present value, depending on the programme.
IDA financing terms and borrowing policy	Debt-distress risk and financing eligibility. SDFP replaced NCBP in July 2020.	Debt risk influences grants versus loans and repayment terms. Under the SDFP, unsatisfactory implementation of policy actions can reduce available allocations, and severe or repeated breaches can also harden financing terms.

Notes: LDCs are least developed countries, LICs low-income countries, LMICs lower-middle-income countries and UMICs upper-middle-income countries. Other ODA eligibility conditions also apply. Under an IMF present-value ceiling, a loan counts at face value when its calculated repayment value is at least that amount.

Sources: OECD DAC (2014), Annex 2; OECD (2019; n.d.-c); IMF (2013; 2021); World Bank (2020; 2024); Chuhan-Pole (2021).

¹IMF financing itself is excluded from these programme debt limits (IMF 2021, 40).

2.4 Related empirical research and this thesis’s contribution

This thesis connects research on sovereign market borrowing costs with the measurement of concessionality in official loans. It draws on studies of market access and borrowing-rate estimation to address gaps in observed rates, and on studies of borrowing-cost differences to interpret patterns across countries. It extends existing comparisons of official creditors by recalculating loan concessionality at borrower-specific discount rates.

ONE’s *Priced Out* analysis compares official lending terms with borrowers’ market financing conditions in 2020–2024. It finds that multilateral loans remained favourable relative to market financing even after their own interest rates rose. The analysis uses the World Bank’s International Debt Statistics (IDS), which provides external-debt data including average interest rates, maturities and grace periods on new commitments. ONE constructs repayment schedules from these terms and reports the present value of repayments per 100 advanced. Its borrower-specific discount rates draw on IDS bondholder terms, secondary-market yields and rating-implied estimates, according to availability. The analysis thus provides a direct precedent for combining different types of borrowing-rate evidence in the measurement of loan concessionality (Harcourt, Haro Ruiz and Rivera 2026; ONE Data n.d.).

The OECD’s *Global Debt Report 2025* examines the market borrowing conditions against which such loans are compared. It documents rising sovereign borrowing costs from 2022, following lower yields in 2021, with differences across income groups, credit ratings and regions. The report uses individual securities to analyse both yields on newly issued bonds and market yields on outstanding bonds. Issue yields describe the terms on which financing was raised, whereas secondary yields describe the market pricing of existing debt. Its primary averages weight larger issues more heavily, and its secondary-market comparisons by region, income and credit rating report median yields across countries. These analyses provide methodological foundations for constructing rates from bond records and substantive comparisons for this thesis’s examination of borrowing conditions over 2012–2024 (OECD 2025, 118–122 and Annex 3.A).

Observed bond issues nevertheless cover an uneven set of borrowers. In Presbitero and coauthors’ 1995–2014 sample, 52 of 105 developing countries issued an international sovereign bond at least once (Presbitero et al. 2016, section 2). Gelos, Sahay and Sandleris likewise find more frequent participation in international credit markets among larger and richer developing countries. Participation must be distinguished from the underlying ability to borrow. A government may issue no bond because it uses other financing or has little financing need, but it may also face unacceptable market terms (Gelos, Sahay and Sandleris 2011, 244–246). An issuance-only comparison can therefore omit countries facing particularly difficult conditions, without the absence of issuance alone establishing that they were unable to borrow. Secondary yields provide additional observations for countries with outstanding bonds during years without new issues (OECD 2025, 121–122). For countries without traded bonds, borrowing-cost estimates instead have to draw on other information about financing conditions and borrower characteristics.

Gueye and Sy (2010) address this information constraint through econometric prediction. They first estimate sovereign ratings from macroeconomic characteristics using an ordered probit model, which

accommodates the ordering of credit-rating categories. They then fit a regression relating observed sovereign spreads—the additional yield above a reference rate—to ratings and global financial conditions. Actual or estimated ratings enter this relationship to predict a country’s spread, which is added to a US Treasury reference rate. For September 2009, the predicted average borrowing rate for their sub-Saharan African sample is 9.60%, compared with 6.68% for the emerging-market benchmark. Excluding distressed Seychelles reduces the predicted African average to 8.34%. These are estimates of borrowing costs conditional on market access, including countries without observed bond yields (Gueye and Sy 2010, 3–4, 8, 15–17).

Gueye and Sy estimate the coefficients linking country characteristics and global conditions to ratings and borrowing spreads. The rating-implied approach used by ONE and this thesis instead assigns a spread associated with the borrower’s rating category and adds a US Treasury rate (ONE Data n.d.). This thesis’s peer estimates use the median rate of other countries selected by specified income, regional and creditworthiness criteria, with creditworthiness partly assessed through a scorecard when actual grades are unavailable. These approaches all use information beyond a country’s own bond transactions, but a fitted regression, a rating-to-spread mapping and a peer comparison derive the estimate in different ways.

The economic information reflected in borrowing rates also matters for their use in concessionality comparisons. Gbohoui, Ouedraogo and Some (2023) investigate whether higher borrowing costs for sub-Saharan African countries remain after accounting for economic and institutional characteristics. Their study examines coupon rates at issuance and spreads on outstanding bonds as distinct outcomes. An additional African difference remains in their initial regressions after conventional risk controls. In expanded spread regressions including financial development, budget transparency, informality and regulatory quality, the regional coefficient is no longer statistically significant. The controls and available sample change, so these results describe different conditional comparisons (Gbohoui, Ouedraogo and Some 2023, 18–20). The policy framework discussed above assigns the same discount rate to borrowers in the same recipient category, although their economic and institutional conditions can differ substantially. This thesis uses borrower-specific rates to incorporate observed or estimated market borrowing-cost differences into official-loan valuations, and examines regional patterns to identify which groups of borrowers face particularly favourable or unfavourable financing alternatives.

Morris, Parks and Gardner (2020) identify borrower-specific discounting as an extension of their comparison of Chinese, IDA and IBRD lending during 2000–2014. They find less favourable Chinese terms and lower concessionality on average in their sample, calculating grant elements with a common 5% discount rate. The authors recognize that countries face different market borrowing rates, but retain the common reference because constructing borrower-specific rates across their full sample would increase complexity and could restrict the comparison to a selected group of countries (Morris, Parks and Gardner 2020, 8–11, 13–14). This thesis applies borrower-specific discounting to usable original loan records from the overlapping 2012–2014 period. It values the same modeled repayment schedules at 5% and at borrower-specific rates to assess the valuation change on those matched loans.

The OECD Creditor Reporting System (CRS) supports a broader application across providers and recipients. Its provider-reported financing records identify recipients, amounts and available loan terms (OECD 2024). Usable CRS records form this thesis’s main loan application because they provide its broadest creditor and geographical coverage. AidData’s later Chinese lending records extend through 2023 and already include grant elements calculated under institutional conventions. The African Debt Database supplies official-loan records within a wider dataset covering domestic and external debt, extending the comparison across African borrowers and creditor portfolios. These applications partly share loans and sources, including the African Debt Database’s use of CRS and AidData. Their differing coverage permits comparisons across creditors, periods and settings, although agreement between results is not independent evidence when the underlying loans overlap (AidData 2025, 28–29; Manger et al. 2025, 2–4, 11–12).

This thesis combines and extends these approaches through an annual borrower-specific dataset for 2012–2024. It retains the different rate sources separately, examines their coverage and agreement, and compares their approximation of observed issuance rates with that of policy references on matching country-years. The evaluation distinguishes the accuracy of annual levels from the reliability of measured changes over time. The dataset also supports analysis of borrowing conditions across countries and years, including regional and income differences. In the loan application, this thesis holds each repayment schedule fixed and changes the discount rate to measure how borrower-specific information alters concessionality relative to policy references. CRS provides the main application, supplemented by the original Morris–Parks–Gardner records and later loan datasets.

3 Constructing the sovereign market-benchmark dataset

3.1 Scope and contents of the dataset

This thesis constructs a country-year dataset of sovereign borrowing benchmarks for 2012–2024. Each record represents one economy in one calendar year. Newly issued bonds provide evidence of borrowing conditions at issuance, but an issuance-based rate requires both an eligible bond issue and sufficiently complete information about its terms. The dataset therefore combines issuance yields with yields on outstanding bonds, reported interest rates on new external debt commitments, and estimates based on sovereign ratings or comparable borrowers. This structure supports the evaluation of benchmark coverage and comparability, followed by an assessment of how borrower-specific discount rates change the measured concessionality of official loans.

The dataset follows a fixed roster of 211 economies for all thirteen years, producing 2,743 country-year records even when no borrowing benchmark is available. The roster contains the non-aggregate entries with a recorded capital city in the archived World Bank country metadata, including territories that satisfy that rule (World Bank n.d.-d). The coverage denominator includes the 1,756 country-years classified as low, lower-middle or upper-middle income in the historical series. High-income and unclassified years remain in the dataset but are excluded from that denominator; missing borrowing-rate observations

remain included. Income groups are drawn from the World Bank’s historical classification workbook and supplemented with World Bank classifications processed by Our World in Data (World Bank n.d.-b; World Bank 2025). Classifications are assigned to the year of the income data on which they are based. For example, the group attached to 2024 reflects the classification published in July 2025 using 2024 income data. This provides a retrospective grouping for the analysis, including the income criterion used to identify comparable borrowers. It does not reproduce the classifications already in force during each year.

The following table summarises the five benchmark sources and estimation methods. Primary and secondary bond yields are constructed separately for US dollars and euros. The main reference selection uses dollar primary and secondary bond yields and dollar-based rating estimates. The International Debt Statistics (IDS) Bondholders series provides an additional proxy based on reported contractual interest rates; its aggregate does not identify the underlying currency composition. Each source-specific rate remains separately identifiable, together with its source, relevant dates and recorded limitations. Where an application requires one benchmark per country-year, a stated selection order determines which eligible reference is used. The alternatives remain available for comparison.

Table 2: Sources and estimation methods for sovereign borrowing benchmarks

Benchmark	Underlying information	Rate and aggregation	Timing and currency
Primary issuance	LSEG records of sovereign bond issues	Mean issue yield, weighted by the amount issued	Issues during the calendar year; separate dollar and euro series
IDS Bondholders	World Bank reports of new public and publicly guaranteed external debt commitments to bondholders	Mean contractual interest rate, weighted by commitment amount	New commitments during the year; currency composition unidentified in the aggregate
Secondary market	LSEG quotations for outstanding sovereign bonds	Mean bond yield, weighted by the recorded face amount outstanding	Eligible quotations closest to year-end; separate dollar and euro series
Rating-implied estimate	Moody’s sovereign ratings, Damodaran’s archived sovereign default spreads and the seven-year US Treasury yield from FRED	Treasury yield plus the sovereign default spread associated with the borrower’s rating	Rating at 1 January, annual mean Treasury yield and year-specific spread archive; dollar reference
Peer approximation	One eligible primary, IDS, secondary or rating-implied rate per comparison country, in that priority order	Median rate of at least three other countries in the first qualifying comparison group	Same-year rates; groups use income, region and actual or estimated creditworthiness. Currency coverage follows the contributing sources

Sources and notes: Author’s compilation and calculations from the sources listed, including archived LSEG extracts (LSEG 2026). Moody’s ratings are drawn from Bloomberg rating-change records (Bloomberg 2026). The IDS definition and amount weighting follow the World Bank’s indicator meta-data (World Bank n.d.-a, “Short definition”). Peer rates approximate borrowing conditions using other countries’ observations and therefore provide a less borrower-specific basis than evidence drawn from the country itself. The following subsections explain the eligibility rules, timing conventions, estimation procedures and selection order.

3.2 Observed bond evidence

The primary and secondary benchmarks are constructed from LSEG bond records, which link issue terms and instrument characteristics to dated yield and price observations (LSEG 2026). The collection and reconciliation of individual bond records build on the OECD’s *Global Debt Report 2025* (OECD 2025, Annex 3.A, pp. 134–136). The OECD provides the security-level collection precedent and the secondary maturity range cited below. The precise annual sampling window, reconciliation thresholds and source-selection rules are this thesis’s implementation choices.

The primary series uses the reported original yield to maturity to account for the issue price and the timing of promised interest and principal payments. Using the coupon rate alone would omit the effect of issuing above or below face value. The main analysis uses the USD bond benchmarks; EUR yields are retained separately because the two currencies have different underlying interest-rate curves. Both currency series require an original maturity of at least one year and a recorded face amount issued equivalent to at least USD 50 million. Provisions allowing early redemption by the issuer (issuer calls) or the retirement of principal before final maturity (sinking funds) can alter repayment timing. The primary series includes otherwise eligible fixed-rate bonds with these provisions when their original issue yield is reported, using that reported yield directly without independently valuing those repayment features. To focus on new medium- and long-term central-government borrowing on fixed-rate terms, the main primary aggregate excludes Treasury bills, central-bank issuance, debt-exchange or restructuring issues, and floating-rate, index-linked and other nonstandard structures.

Before aggregation, repeated records are reconciled using the borrower, issue and maturity dates, currency, coupon and recorded issue amount. Records representing the same underlying issue contribute only one issuance weight, using the median of their finite reported original yields. If their reported yields differ by more than 0.25 percentage points and the discrepancy remains unresolved, that issue is excluded from the average. Reopenings or tranches that cannot confidently be identified as the same issue remain separate. For the dollar series, the country-year primary rate is

$$r_{ct}^P = \frac{\sum_{i \in I_{ct}} A_i y_i}{\sum_{i \in I_{ct}} A_i}, \quad (2)$$

where I_{ct} contains the eligible issues of country c in year t , y_i is the original issue yield in percent, and A_i is the recorded dollar face amount issued. The euro series is calculated separately using the source’s dollar-equivalent issue amounts. The weighting describes the average yield on the financing volume represented in the sample. The number of issues, their maturities and the largest issue’s weight are retained alongside the rate. A year with only one issue remains usable, but its rate describes that particular borrowing operation. The accompanying results identify years in which one issue accounts for at least 80% of the aggregate weight, because their average is dominated by the terms of that issue.

The secondary series describes borrowing conditions around year-end using quotations: recorded market prices or yields for bonds that have already been issued. Its construction has two steps: obtain a yield to maturity for each eligible bond, then average those yields into a country-level rate. For a given country,

year and currency, the benchmark averages the eligible yields reported by LSEG. Only when no eligible reported yields are available is the benchmark calculated from bond prices instead. Price-derived yields are therefore not added to an average already based on reported yields.

To calculate a yield from a price, the bond's remaining coupon and principal payments are discounted at the rate that makes their present value equal to its current price, including accrued interest. This defines yield to maturity (FINRA 2022). The implementation uses the reported mid-price, treated as a clean price excluding interest accrued since the preceding coupon payment. Adding that accrued interest gives the price used in the calculation, following the standard present-value relationship set out by Djatschenko (2020, sec. 2.4, eq. 1):

$$P_i^{\text{clean}} + AI_i = \sum_{j=1}^{N_i} \frac{CF_{ij}}{(1 + y_i/(100f_i))^{f_i\tau_{ij}}}. \quad (3)$$

Here, prices, accrued interest AI_i and payments CF_{ij} are expressed per 100 units of face value. The payments comprise the remaining coupons and principal repayment of 100 at maturity. There are N_i remaining payment dates; f_i is the number of coupon payments per year. The time from the quotation date to payment j , τ_{ij} , is measured in years as actual days divided by 365.25. The yield y_i is a nominal annual percentage compounded at the coupon frequency and is solved numerically. Yields enter the country averages in their quoted form, without conversion to a common effective annual convention. This calculation is used only where the recorded prices and instrument terms support a fixed-coupon repayment schedule. Coupon dates are reconstructed backwards from maturity, and accrued interest is approximated from the elapsed fraction of the regular coupon period. The quotation date serves as the valuation date.

For each quoted instrument and year, the date rule selects the reported yield or complete price quotation closest to 31 December within a window extending 31 calendar days before and seven days after that date. The earlier observation is chosen in a tie. For the secondary benchmark, this thesis restricts remaining maturity to 2–15 years, matching the maturity range used in the OECD's secondary-market comparison (OECD 2025, fig. 3.7, panel A, p. 121). The restriction limits differences arising from combining bonds close to repayment with much longer-term debt. Remaining maturity is measured at year-end for reported yields and at the quotation date for yields calculated from prices. To make yields more comparable across bonds with different contractual features, the secondary series excludes bonds with early-redemption, principal-retirement or conversion provisions, including callable, putable, sinking-fund and convertible bonds, as well as other nonstandard instruments. Its instrument scope is therefore narrower than that of the primary series.

Both secondary routes weight the eligible bond yields by recorded face amounts outstanding. For a given currency, the resulting country-year rate is

$$r_{ct}^S = \frac{\sum_{i \in J_{ct}} W_i y_i}{\sum_{i \in J_{ct}} W_i}, \quad (4)$$

where J_{ct} contains the eligible issues from the selected route and W_i is the dollar-equivalent face amount outstanding recorded in the archived LSEG extracts acquired in 2026. These instrument attributes are reused with historical quotations, and their own observation dates are not retained. The year attached to a quotation therefore does not date its outstanding-balance weight. Multiple identifiers representing the same issue are consolidated before weighting. The secondary average therefore uses recorded amounts rather than reconstructed historical year-end balances. Differences from the primary rate can reflect both changes in borrowing conditions and differences in the instruments and weights represented.

3.3 IDS bondholder borrowing terms

The World Bank’s IDS Bondholders data and the primary series cover overlapping bond borrowing, but measure its terms differently. The primary series combines yields on individual eligible issues; IDS reports the average contractual interest rate on new public and publicly guaranteed external debt commitments to bondholders in each country and year. The World Bank weights these interest rates by the amounts committed (World Bank n.d.-a). Unlike an issue yield, a contractual interest rate does not capture the effect of issuing above or below face value. Using IDS bondholder rates as a borrower-specific reference builds on ONE’s *Priced Out* methodology, which combines them with secondary-market yields and rating-based estimates (ONE Data n.d., “The discount rate”).

IDS can provide broader coverage because it includes borrowing by other public bodies and private borrowers with public guarantees (World Bank n.d.-a). The country-year Bondholders aggregate also combines commitments whose currency composition is not identified in the series used here. IDS is therefore retained as a distinct borrowing-rate proxy, with broader borrower coverage and without the USD-only restriction applied to the primary bond benchmark. Its agreement with primary issuance is evaluated with this overlap in underlying borrowing in mind.

The analysis uses positive IDS rates and excludes observations whose interpretation as a borrowing rate remains unresolved.² A figure that may represent only a floating-rate margin is withheld because the margin excludes the reference interest rate needed to calculate the full interest charge.

3.4 Rating-implied borrowing rates

Sovereign credit ratings provide a basis for estimating borrowing rates, including for countries without observed bond yields. The estimate combines a US Treasury reference rate with a default spread associated with the borrower’s rating. The spread represents the additional yield associated with sovereign credit risk. This approach follows ONE’s *Priced Out* methodology, which adds a rating-based spread to the annual average seven-year US Treasury yield (ONE Data n.d., “The discount rate”). For country i in year t , the calculation is

²Eight country-year observations report grace periods longer than their maturities. Their interest rates remain eligible as borrowing-rate proxies, but the inconsistent grace and maturity pairs are not used to construct repayment schedules.

$$\hat{r}_{it}^{\text{rating}} = \bar{y}_t^{\text{US},7} + s_t(R_{i,1 \text{ January } t}), \quad (5)$$

where $\bar{y}_t^{\text{US},7}$ is the annual average seven-year Treasury yield and $s_t(R_{i,1 \text{ January } t})$ is the spread assigned to the country's rating at the beginning of that year. The Treasury yield and resulting estimate are expressed in per cent, and the spread in percentage points.

The Treasury component is the arithmetic mean of the available daily observations in FRED's DGS7 series (Board of Governors of the Federal Reserve System 2026). The seven-year maturity follows ONE's choice of a reference closer to the typical duration of emerging-market sovereign bonds than a ten-year yield (ONE Data n.d., n. 14). It supplies a common dollar reference for the country estimates; it is not matched separately to the maturity of each bond or official loan.

The country component uses the Moody's foreign-currency sovereign rating in force on 1 January, reconstructed from Bloomberg's rating-change records (Bloomberg 2026). That rating label is matched to the corresponding group in Damodaran's historical default-spread tables, whose sovereign grades are based on local-currency ratings (Damodaran n.d.). The calculation therefore transfers spreads between foreign- and local-currency rating categories, which need not coincide for a given country. For each year, the calculation takes the median default spread among country entries carrying the same rating in the relevant annual archive. The median supplies one spread per rating group where individual entries differ. An estimate requires both a Moody's rating and a matching spread; ratings from other agencies do not substitute for a missing Moody's observation.

This construction produces a retrospective annual estimate. The rating is fixed at the start of the year, while the Treasury component reflects observations throughout the year and the spread comes from the historical table assigned to that year.³ Countries with the same rating therefore receive the same estimate within a year. Differences in their borrowing conditions that the rating does not capture remain outside this approximation.

This thesis retains the Treasury-plus-spread formula as the main rating-based reference, without a fitted correction. Section 4.5 evaluates corrections using observed issuance and explains this choice: their gains are strongest when earlier observations are available for the same borrower, whereas most country-years relying on rating estimates lack that history.

3.5 Peer-based borrowing-rate approximations

For countries without a usable borrowing-rate reference from the preceding sources, this thesis estimates a rate from a group of selected countries in the same year. The estimate is the median of their rates, with the group selected according to income category, geographical region and creditworthiness. The IMF–World Bank debt-management manual illustrates using a comparable sovereign's credit spread to

³The historical spread data are assigned to years according to the labels of the annual releases. The observation dates of individual spreads are not consistently stated, and some releases contain updates from the following year. Together with the annual Treasury average, this prevents interpreting the estimates as information available on 1 January.

approximate borrowing costs when direct evidence is unavailable (Balibek et al. 2019, Annex L, 59–60). This thesis applies that principle across the annual dataset through an explicit procedure for selecting comparison countries and combining their rates.

Each comparison country contributes one eligible rate. Its primary issuance rate is used when available, followed by IDS bondholder terms, secondary-market yields and an existing rating-implied rate. The country whose rate is being estimated is excluded, and peer estimates are not used as inputs to other peer estimates. The pool therefore combines observed borrowing information with rating-implied estimates. The differences in timing and currency coverage described for the individual sources also carry into the peer comparison, including the unidentified currency composition of IDS bondholder terms.

Income matching requires the same World Bank income category: low, lower-middle, upper-middle or high income. It uses the retrospective income-data-year classifications described earlier. Regional matching requires membership in the same geographical group. Countries within these categories are not ranked by how close their income levels or geographical locations are. High-income countries can enter groups that do not require an income match, or groups formed for high-income borrowers.

Creditworthiness matching uses the available Moody’s rating at the beginning of the year. When that rating is missing from the dataset, this thesis estimates creditworthiness using the scorecard-based approach implemented by Abate, Brown, Sienaert and Thomas (2021, 4–6 and Appendix D), drawing on Moody’s published sovereign-rating methodology. Abate and coauthors used the scorecard to estimate ratings for developing countries with incomplete actual-rating coverage. This thesis adapts their approach to identify comparable borrowers. The estimated grades are used for matching, rather than to create an additional rating-implied borrowing-rate series.

The scorecard assigns economic indicators to strength categories using published thresholds. For example, average growth above 4.5% receives the strongest growth category, whereas growth below 0.5% receives the weakest. Growth contributes 25% of the economic-strength calculation, with growth variability, competitiveness, economic size and income per person supplying the remainder. Governance and inflation indicators determine institutional strength, and debt and interest-payment indicators determine fiscal strength. Moody’s combination tables translate these factor scores, together with the event-risk assessment, into a grade on its credit-rating scale (Moody’s Investors Service 2018, 4–6, 9, 14 and 18).

This thesis retains governance indicators in the institutional assessment. Abate and coauthors excluded them because their subsequent regressions used related governance measures to explain the estimated ratings, creating a mechanical connection between the variables. That concern does not require their exclusion from the peer-matching calculation. The treatment of event risk follows their simplifying assumption of a moderate level for every country. It therefore does not reproduce each country’s agency assessment of political, liquidity, banking and external vulnerability risks (Abate et al. 2021, 5–6).

When gross interest expenditure is unavailable and the relevant indicator has a positive weight, the calculation allows its score to vary from the strongest to the weakest category. It carries these possibilities, together with bounds on adjustments for default history and foreign-currency debt exposure, through the combination tables to obtain lower and upper grade bounds. These bounds describe uncertainty

within the adopted calculation and remain conditional on its assumptions, including moderate event risk. An available actual rating always takes priority over the estimated range.

The matching procedure permits a creditworthiness difference of at most three rating steps. When an estimated range is used for either country, the most distant pairing within the two ranges must still satisfy that limit. The calculation then selects all members of the first group in the following table containing at least three eligible countries. Each group is assessed against the same pool of available countries. The conditions change between groups, rather than adding countries one by one to the preceding group.

Table 3: Peer-group selection criteria

Search order	Conditions required for inclusion
1	Same income category, same region and creditworthiness within three steps
2	Same income category and creditworthiness within three steps
3	Same region and creditworthiness within three steps
4	Creditworthiness within three steps
5	Same income category and same region

Creditworthiness is assessed using an available actual rating or the estimated grade range. At least three distinct comparison countries are required in every group.

The sequence balances similarity with the availability of comparison rates. The first group matches all three characteristics. Subsequent groups allow particular characteristics to differ when the earlier conditions produce too few countries. The fifth group permits matching on income and region without a creditworthiness condition. If none of the five groups qualifies, no peer estimate is calculated. The matching sequence, three-step limit and minimum of three countries are this thesis’s choices for implementing the comparison, rather than requirements of the cited scorecard or debt-management manual.

The estimate is the unweighted median of all rates in the selected group. Each country contributes once, without additional weights for economic size, number of bonds or closeness within the permitted rating distance. Any weighting used to construct that country’s primary or IDS rate is retained. The median limits the influence of individual extreme rates, and the minimum of three countries reduces reliance on a single comparator. Three is a minimum group size, not a requirement to select exactly three countries or the three closest matches.

During 2012–2024, peer estimates supply the selected reference for 731 low- and middle-income country-years across 89 countries. In 582 of these cases, matching uses an estimated target grade; two use a recorded rating, and 147 use income and region without a creditworthiness restriction. These counts concern cases in which a peer estimate is selected, after the preceding rate sources have been considered. The full low- and middle-income panel contains 903 country-years with recorded ratings out of 1,756. The annual distribution of the selected peer references across matching groups is shown below.

Table 4: Selected peer references by matching group, 2012–2024

Year	Income + region + grade	Income + grade	Region + grade	Grade only	Income + region	Total
2012	2	21	5	27	21	76
2013	4	19	14	17	19	73
2014	11	17	11	9	17	65
2015	15	23	2	10	13	63
2016	13	21	8	8	13	63
2017	20	18	5	4	11	58
2018	24	19	1	1	11	56
2019	14	24	1	3	10	52
2020	22	16	1	2	7	48
2021	21	16	1	3	5	46
2022	24	12	1	3	4	44
2023	20	17	0	1	5	43
2024	17	12	1	3	11	44
2012–2024	207	235	51	91	147	731

Counts are low- and middle-income borrower-years using a peer rate as their selected reference. Grade matches use actual ratings or estimated ranges under the three-step condition. Source: this thesis's calculations.

The selected groups contain an average of 6.72 countries and a median of five. The middle half contains four to eight countries. Group sizes range from three to 26, with a standard deviation of 4.50. These statistics describe the group used for each borrower-year, so a group shared by several borrowers is counted for each use.

The following table shows the rate sources contributing to those groups. Across the 731 selected borrower-years, there are 4,915 contributions from comparison countries. A country appearing in several borrowers' groups contributes separately to each. Rating-implied rates account for 2,867 of these contributions and enter 697 selected groups, including 71 composed entirely of rating-implied rates. Their use is therefore a substantial part of the peer estimates' empirical basis.

Table 5: Rate sources contributing to selected peer groups, 2012–2024

Year	Selected borrower-years	Primary	IDS	Secondary	Rating- implied	Total contributions
2012	76	84	212	0	378	674
2013	73	22	188	0	288	498
2014	65	55	126	2	288	471
2015	63	55	141	21	281	498
2016	63	40	112	22	302	476
2017	58	82	34	12	182	310
2018	56	79	59	18	151	307
2019	52	56	55	32	121	264
2020	48	51	36	26	176	289
2021	46	56	44	24	149	273
2022	44	42	13	30	186	271
2023	43	29	21	43	219	312
2024	44	82	16	28	146	272
2012–2024	731	733	1,057	258	2,867	4,915

Each contribution is one comparison country's rate entering one target country's peer group. Repeated use of the same country-year is counted separately. The source counts sum to contributions, not to selected borrower-years or to additive effects on the median. Source: this thesis's calculations.

Shared comparison countries and rating-implied inputs mean that different peer estimates can depend on the same underlying information. The dataset retains the member countries, their rate sources and the dispersion of their rates for examining particular comparisons. The benchmark-evaluation chapter assesses accuracy against observed issuance rates, excluding the target country from its own peer group. It separately examines matching with an available target rating and matching when that rating is withheld to assess the use of estimated creditworthiness. Because these tests require observed issuance, they provide less direct evidence about accuracy for countries without bond-market observations.

3.6 Selecting a borrowing-rate reference

The dataset retains primary issuance rates, IDS bondholder terms, secondary-market yields, rating-implied rates and peer approximations separately. When an application requires one borrowing-rate reference for a country-year, this thesis selects the first eligible rate in that order. Eligibility depends on the source-specific construction requirements and the suitability of the observation for the borrowing-cost comparison. Each selected rate retains its source, timing and currency information.

Primary issue yields receive priority because they describe the terms on which the sovereign raised financing. IDS bondholder terms come next because they also concern new borrowing during the year. The two series cover overlapping bond borrowing, although IDS reports contractual interest rates rather than issue yields. Its public and publicly guaranteed debt coverage can include borrowers beyond the central government, and the country-year series used here does not identify the currencies of the underlying commitments (World Bank n.d.-a). These differences affect comparability without making IDS a separate, independent observation of the same borrowing conditions.

After primary issuance rates and IDS terms, the selection uses secondary yields, which describe the market pricing of outstanding bonds around year-end and can supply observations in years without new issuance. If none of these sources provides an eligible rate, the selection proceeds to the rating-implied estimate and then to the peer approximation. This thesis chooses the sequence to prioritize evidence on new borrowing, followed by market pricing and estimates. The benchmark-evaluation chapter assesses how closely the different sources approximate observed issuance rates and compares the main selection with an alternative that places secondary yields before IDS. The prescribed order does not itself establish an accuracy ranking.

For a floating-rate loan, the interest charged consists of a reference interest rate plus an additional margin. A reported figure that may contain only that margin cannot establish the total borrowing cost and is withheld if its interpretation remains unresolved. Distress and restructuring can likewise affect whether a bond yield represents the cost of new borrowing on ordinary contractual terms. Exclusion depends on the evidence and intended use, rather than a high rate alone. When the country's circumstances prevent

using an ordinary borrowing-cost estimate, the procedure does not replace an excluded observation automatically with a rating-implied or peer rate.

If no source supplies an eligible reference, the country-year remains in the dataset without a selected rate. Missing source information, the absence of observed issuance and evidence of financing difficulties remain distinguishable. The absence of an observed bond issue does not by itself establish that a country was unable to borrow.

The benchmark evaluation and concessionality applications also use selections that exclude peer approximations or restrict the evidence to primary issuance and IDS. The main official-loan applications exclude peer-only references and report the broader selection separately. Because these restrictions can change both the rate and the sample, differences between the resulting estimates are interpreted with reference to the loans and countries included in each comparison.

When a country’s selected source changes between years from an annual new-commitment rate to a year-end secondary yield, the measured movement may reflect the different measurement dates as well as changing borrowing conditions. Even within the same year, secondary yields observed in December can differ from yields on bonds issued earlier because market conditions changed in the intervening months. The analysis of borrowing conditions across countries and over time therefore examines individual sources and comparisons that hold the method fixed alongside the selected series.

3.7 Construction example and chapter close

For Peru in 2024, the primary borrowing-rate reference combines two USD bond issues dated 8 August. Weighting their issue yields by the amounts raised gives 5.70775%, rounded to 5.71%. This primary rate supplies the selected reference under the source order described above.

Table 6: Construction of Peru’s 2024 primary borrowing-rate reference

Maturity date	Amount issued (USD billion)	Issue yield (%)	Weight (%)
8 February 2035	1.25	5.384	41.67
8 August 2054	1.75	5.939	58.33

Source: LSEG (2026); weights calculated by this thesis. Displayed weights are rounded.

Reproducing the vendor-based calculations requires access to the licensed source data. The following chapter evaluates the dataset’s coverage, compares rates available for the same country-year and assesses how closely alternative references approximate observed primary issuance rates.

4 Evaluating the borrowing-rate benchmarks

4.1 Evaluation design

This chapter evaluates how widely the dataset represents low- and middle-income countries over 2012–2024 and how closely its borrowing-rate estimates approximate observed sovereign issuance. Coverage is assessed across all eligible country-years. The accuracy comparisons use the smaller sample with an eligible US-dollar primary issuance rate, comparing IDS bondholder terms, secondary yields, rating-implied rates and peer estimates separately with primary issuance for the same country and year. Primary rates describe financing actually raised during the year, although differences between sources also reflect their instruments and measurement dates. Because IDS and primary rates partly describe the same underlying bond borrowing, their agreement reflects consistency between overlapping sources rather than independent evidence of predictive accuracy.

Each comparison with a policy rate includes only country-years for which the primary rate, the alternative borrowing rate and the applicable DAC reference are all available. The alternative and policy rate are therefore evaluated against exactly the same primary observations. Since availability differs between methods, their separate comparisons with the policy rate can involve different countries and years. Direct comparisons between two methods instead use the observations for which both methods and primary issuance are available. Each matched country-year receives equal weight in the error summaries. The amount weights used to construct primary rates from individual bond issues do not carry over into these comparisons across country-years. Supplementary summaries give each country, or each year, equal total weight to assess how the uneven number of observations affects the results. Sample counts distinguish observations, countries and years.

The main accuracy comparison asks whether borrower-specific rates approximate observed primary issuance more closely than the DAC discount rates of 9%, 7% and 6%. This tests their closer correspondence to issuance, not the adequacy of DAC rates for their separate accounting purpose. The comparison is retrospective and does not measure forecasting performance using only information available before issuance. It focuses on 2018–2024 because the grant-equivalent system using these differentiated rates became the headline basis for ODA reporting for 2018 flows, following transitional reporting in 2015–2017 (OECD DAC 2014, Annex 2, paragraphs 5–10; OECD 2019). This thesis assigns the applicable rate according to each country’s DAC recipient category in the relevant year. These categories follow the historical DAC recipient lists, separately from the retrospective World Bank income groups used to describe coverage and select peers. To assess performance over the longer period, the same category-based rule is applied over 2012–2024 and separately over 2012–2017. Its application before 2018 provides a standardized analytical comparison. A separate calculation for 2012–2017 uses the historical 10% convention described in Section 2.3.

For each observation, the error is the alternative rate minus the primary rate, expressed in interest-rate percentage points. The mean signed error indicates whether an alternative lies above or below observed issuance on average, but positive and negative discrepancies can offset one another. This thesis therefore

uses mean absolute error (MAE) as its principal accuracy measure, expressing the average distance from primary issuance in percentage points. Root mean squared error (RMSE) gives greater influence to large discrepancies and accompanies MAE to show whether these materially affect the assessment (Hyndman and Athanasopoulos 2021, section 5.8).

The improvement over the policy reference is calculated for each country-year by subtracting the alternative’s absolute error from the policy rate’s absolute error. The average of these paired differences equals the reduction in MAE on the matched sample. A positive value means that the borrower-specific alternative is closer to observed issuance on average. Direct comparisons between two borrowing-rate methods use the same calculation on their shared observations.

To assess uncertainty around these average improvements, this thesis repeatedly draws samples of countries, keeping each country’s observed years together. Retaining whole country histories preserves the relationship between observations of the same borrower across years. This follows the pairs cluster-bootstrap approach described by Cameron and Miller (2015, section II.F). Each of 4,999 draws samples the original number of countries with replacement and divides the sum of their loss differences by their total observation count. Approximate 95% intervals use the 2.5th and 97.5th percentiles of these means, with R’s type-7 quantiles.⁴

Shared market conditions and common inputs to estimated rates can also produce related movements across countries in the same year. Country-based resampling does not capture this relationship. A second calculation fits an intercept-only model to the paired loss differences and uses a country-and-year clustered covariance estimate, with HC1 scaling, finite-cluster corrections and the singleton-intersection correction. Tests use a Student’s t distribution with degrees of freedom equal to the smaller of the country and year counts minus one, giving six in the 2018–2024 comparison. Only seven calendar years provide limited information about movements shared across countries (Cameron and Miller 2015, sections V–VI; MacKinnon, Nielsen and Webb 2023). This thesis consequently treats these uncertainty estimates as exploratory and places greater emphasis on the size of the observed accuracy improvements and their stability across the alternative comparisons.

4.2 Coverage and the sources of available rates

The dataset supplies a selected borrowing-rate reference for 1,701 of the 1,756 low- and middle-income country-years in the sample, giving overall coverage of 96.9%. The sample contains 142 economies over 2012–2024. Each economy enters the sample in the years when it belongs to a low- or middle-income group under the retrospective classification described in Chapter 3, so the number of countries represented varies across years.

⁴For two-sided tests, each country’s summed loss difference is centred by subtracting its observation count times the full-sample mean. Resampled centred means give $p = (1 + B_{\geq})/5000$, where $B_{\geq}$ counts absolute values at least as large as the observed mean’s absolute value. Multiple comparisons use Holm adjustment within each reported family (Holm 1979).

Primary issuance alone provides a reference for 215 country-years, or 12.2% of the sample. Adding IDS bondholder terms and secondary yields extends coverage to 516 country-years, or 29.4%. Rating-implied rates supply another 454 observations, bringing coverage to 55.2%, and peer estimates add 731. Estimates therefore account for a substantial part of the dataset’s reach beyond countries and years with usable issuance, IDS or secondary-market information.

Table 7: Availability and contribution of each rate source, 2012–2024

Source or estimation method	Country-years with an eligible rate	Additional country-years covered in selection order	Cumulative coverage
Primary issuance	215	215	215 (12.2%)
IDS bondholder terms	418	229	444 (25.3%)
Secondary yields	251	72	516 (29.4%)
Rating-implied rates	893	454	970 (55.2%)
Peer estimates	1,698	731	1,701 (96.9%)

Notes: All percentages use the denominator of 1,756 low- and middle-income country-years. Eligible rates satisfy the requirements described in Chapter 3. The additional contribution follows the selection order shown in the table. *Source:* Calculations from the dataset constructed in Chapter 3.

Because several sources can provide a rate for the same country-year, the eligible counts overlap and cannot be added. The additional contributions count each country-year once under the selection order. Unselected rates remain available for comparisons between sources.

The share of country-years using a peer reference falls from 55.1% in 2012 to 33.1% in 2023 and 34.4% in 2024. Between 2012 and 2024, coverage without peers rises from 39.1% to 56.3%. Between 2021 and 2022, the number with a primary issuance rate, IDS terms or a secondary yield falls from 50 to 37, but the number with any selected rate falls only from 132 to 129. Rating-implied and peer estimates thus maintain broad coverage despite the reduction in observations from those three sources.

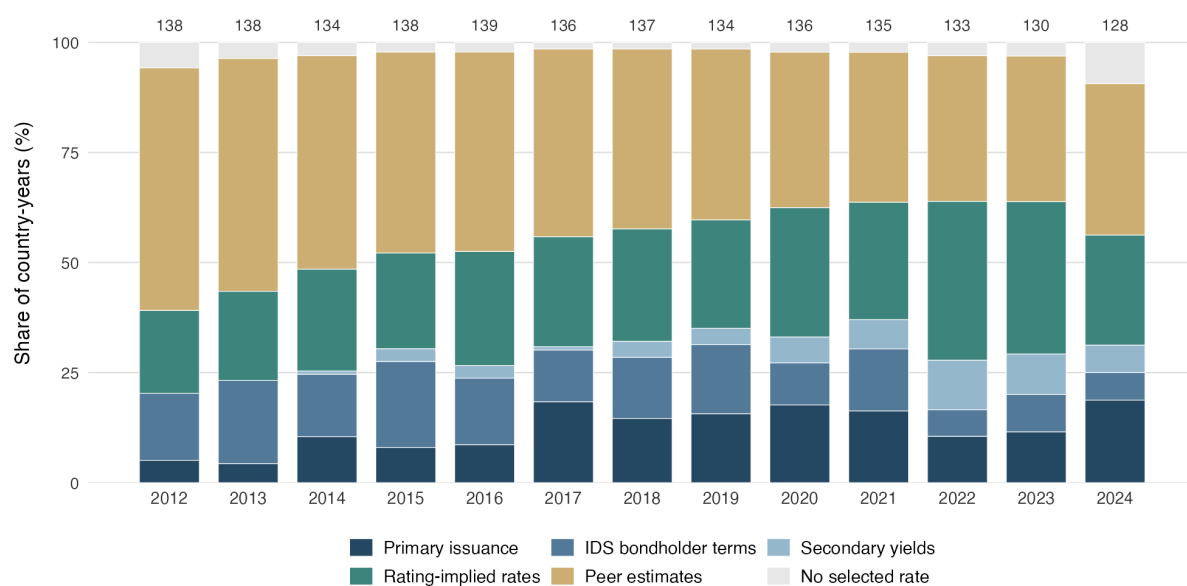


Figure 1: Source of the selected borrowing-rate reference by year. Selected borrowing-rate sources as shares of the low- and middle-income sample in each year, 2012–2024. Numbers above bars show annual sample sizes.

Notes: Each bar covers all low- and middle-income country-years in that year, including those without a selected rate. Numbers above the bars give the annual sample size. Eight of the twelve observations without a selected rate in 2024 are excluded because distress, restructuring or related circumstances prevent using their rates as ordinary borrowing-cost references. *Source:* Calculations from the dataset constructed in Chapter 3.

Low-income countries depend much more heavily on estimates than the two middle-income groups. Primary issuance, IDS or secondary yields cover only 22 of 389 low-income country-years (5.7%), compared with 32.0% of lower-middle-income observations and 39.9% of upper-middle-income observations. Adding rating-implied rates raises coverage without peers to 19.8%, 56.8% and 73.1%, respectively. Peer estimates supply the selected reference in 71.0% of all low-income country-years, against 41.7% in the lower-middle-income group and 25.7% in the upper-middle-income group.

Regional coverage likewise differs in the information supporting the selected rates. In sub-Saharan Africa, primary, IDS or secondary evidence covers 16.0% of country-years, and peer estimates are selected in 57.1% of all country-years. In Latin America and the Caribbean, the corresponding shares are 49.5% and 19.1%. Comparisons of borrowing conditions across these regions therefore draw to different degrees on observed borrowing information and estimates.

The dataset retains 55 country-years without a selected rate: 47 lack a reference meeting the source requirements, and eight are excluded because distress, restructuring or related circumstances prevent an ordinary borrowing-cost interpretation. An absent reference consequently cannot be read as evidence that a country was unable to borrow.

Accuracy comparisons use the smaller sample with observed primary rates, including only five low-income country-years before further matching requirements. Section 4.5 examines how far that evidence supports using estimated rates for countries without observed issuance.

4.3 Approximation accuracy relative to policy rates

This section examines whether borrower-specific rates approximate observed primary issuance more closely than the differentiated DAC discount rates. During 2018–2024, all four alternatives—IDS bondholder terms, secondary yields, rating-implied rates and peer estimates—have lower mean absolute error than the DAC rates on their respective matched samples.

For secondary yields, rating-implied rates and peer estimates, the reduction in mean absolute error is approximately 30–34%, equivalent to 0.50–0.58 interest-rate percentage points. The larger reduction for IDS partly reflects its overlap with primary issuance in the underlying bond borrowing. Close agreement between these two sources therefore provides less independent evidence about approximation accuracy than the results for the estimated rates.

Table 8: Approximation of primary issuance rates relative to DAC references, 2018–2024

Alternative reference	Country-years	Countries	Alternative MAE	DAC MAE	Reduction in MAE
IDS bondholder terms	127	37	0.573	1.654	65.4%
Secondary yields	117	32	1.110	1.662	33.2%
Rating-implied rates	126	38	1.108	1.686	34.3%
Peer estimates	136	41	1.161	1.661	30.1%

Notes: MAE is in interest-rate percentage points. Each row compares the alternative and the DAC rate against the same primary observations. The samples differ between rows, so the table does not establish an accuracy ranking between methods. *Source:* Calculations from the constructed dataset using the evaluation design in Section 4.1.

The DAC rates exceed primary issuance rates on average by 0.43–0.63 percentage points across these matched samples. Among the alternatives, secondary yields are lower than primary issuance by 0.34 points on average, while rating-implied and peer rates are higher by 0.10 and 0.27 points, respectively. The IDS difference is close to zero, at -0.02 points. The small average difference for rating-implied rates nevertheless coexists with an MAE above one percentage point, as overestimates and underestimates offset one another in the signed average.

All four alternatives also reduce RMSE relative to their DAC comparison, although the improvement for secondary yields is much smaller under this measure. Their RMSE is 1.98 percentage points, against 2.12 for DAC. The improvement is therefore less pronounced when larger errors receive greater weight. Rating-implied and peer rates have RMSEs of 1.45 and 1.50 points, against DAC values of approximately 2.1 in their respective samples.

Giving each country, or each year, equal total weight leaves the reduction in mean absolute error positive for all four alternatives during 2018–2024. Under the country-bootstrap procedure described in Section 4.1, the 95% confidence intervals for these reductions are above zero for all four alternatives. The corresponding tests remain statistically significant at the 5% level after applying Holm’s (1979) correction for the four comparisons.

Accounting additionally for related observations across countries in the same year weakens that statistical evidence. Only IDS remains statistically significant at the 5% level after the same multiple-comparison correction, although the estimated average reductions in error are unchanged. With only seven calendar years, the evidence for improvement across a broader range of borrowing conditions remains limited, as discussed in Section 4.1.

Over the full 2012–2024 period, all four alternatives have lower MAE than the standardized 9/7/6% reference. The earlier years are less uniform. During 2012–2017, IDS, secondary yields and peer estimates improve on that reference, but rating-implied rates have a slightly higher MAE, at 1.26 rather than 1.16 percentage points. Their better approximation during 2018–2024 therefore does not hold throughout the study period. Against the historical 10% convention, all four alternatives show much larger reductions of 3.01–3.73 points in 2012–2017. Because these two earlier-period comparisons retain exactly the same

observations and borrowing-rate estimates, the larger gains against 10% result from changing the policy reference.

Rating-implied rates reproduce the ordering of countries from lower to higher observed borrowing costs more closely than peer estimates in every year included in this comparison. This thesis measures that agreement using Spearman rank correlations with primary issuance, calculated on the same countries for both methods within each year. The summary includes years with at least eight matched countries, a reporting threshold chosen for this comparison, leaving 189 country-years over 2014–2024. The equally weighted average of the eleven annual correlations is 0.78 for rating-implied rates and 0.60 for peer estimates. Both methods distinguish relatively higher- and lower-cost issuers within this sample. This finding concerns relative positions across countries, separately from the size of errors in their estimated rates.

4.4 Source comparability and differences between methods

This section examines whether some sources approximate observed issuance more closely than others, and why the sources can nevertheless give different rates for the same country and year. It also assesses how these differences affect the choice of a borrower-specific reference rate.

Table 9: Paired comparisons of approximation accuracy, 2018–2024

Methods compared	Country-years	Countries	First: MAE	Second: MAE	First: median	Second: median
Secondary / rating-implied	107	29	1.115	1.093	0.769	0.937
Rating-implied / peer	126	38	1.108	1.136	0.936	0.875
Secondary / peer	117	32	1.110	1.131	0.779	0.865

Notes: All error measures are in interest-rate percentage points relative to primary issuance. “Median” denotes the median absolute error, which is less sensitive to unusually large discrepancies than MAE. “First” and “second” follow the order of methods named in each row. Both methods use identical observations within a row, but samples differ between rows. *Source:* Calculations from the constructed dataset using the evaluation design in Section 4.1.

None of the three paired MAE comparisons establishes an accuracy advantage significant at the 5% level under the country-bootstrap procedure. For secondary minus rating-implied MAE, the 95% interval extends from -0.23 to $+0.35$ percentage points, permitting either method to have the lower average error. Secondary yields nevertheless have lower median absolute errors than either estimate on their respective matched samples. Peer estimates also have a lower median than rating-implied rates, despite a slightly higher MAE. The relative performance therefore depends partly on sensitivity to large discrepancies.

Across 189 overlapping country-years in 41 countries during 2012–2024, IDS rates are 0.13 percentage points below primary issuance on average, with a mean absolute difference of 0.58 points. About 81% of the paired rates differ by no more than one point. Their overlapping bond borrowing helps explain

this agreement, although differences in contractual interest versus issue yields and in the commitments covered prevent treating them as interchangeable observations, as discussed in Chapter 3.

Across 150 overlapping country-years in 36 countries during 2012–2024, secondary yields are 0.37 percentage points below primary issuance on average, with a mean absolute difference of 1.02 points. These comparisons use all eligible source overlaps, without requiring a DAC reference. Primary rates describe financing raised during the year, whereas secondary yields describe outstanding bonds around year-end. Changes in market conditions between those dates and differences in maturity can therefore contribute to the discrepancies.

In the 17 country-years with both relatively close observation dates and similar aggregate maturities, the mean absolute rate difference is 0.50 points.⁵ This smaller gap does not isolate the contribution of timing or maturity, because the countries and bonds also differ. Closer dates alone do not consistently coincide with smaller discrepancies.

The two largest primary–secondary gaps account for approximately 60% of the sum of squared differences. Omitting them reduces RMSE from 1.81 to 1.14 points, compared with a smaller fall in MAE from 1.02 to 0.90 points. The full sample remains the main comparison, since the size of a discrepancy alone does not establish that an observation is erroneous.

The selection order in Chapter 3 consequently rests on the nature and timing of the borrowing evidence as well as approximation accuracy. Reversing the order of IDS and secondary yields changes the selected rate in 29 country-years across 12 countries, without changing the total coverage of 1,701 country-years. Among those affected observations, the mean absolute change is 2.60 percentage points. The order therefore affects relatively few selections but can substantially change the reference rate in those cases.

4.5 Applicability of estimated rates and the calibration evidence

This section examines how far accuracy among observed bond issuers supports the use of estimated rates for other borrowers. It also assesses whether corrections fitted to observed issuance should replace the original rating-implied formula. The evaluation samples differ substantially from the country-years in which estimates supply the selected reference. Low-income borrowers account for only 5 of 214 country-years in the peer evaluation over 2012–2024, compared with 276 of the 731 country-years in which peer rates are selected. Their borrowing conditions are therefore sparsely represented in the direct accuracy evidence.

This thesis estimates a common correction by regressing observed issuance rates on the original rating-implied rates, with an intercept and slope. A further adjustment uses the country’s earlier discrepancies

⁵This thesis defines close dates for this diagnostic as an interval of 0–90 days from the last primary issue to the earliest secondary observation, and similar maturities as a difference of no more than three years between the aggregate maturities. These cutoffs do not change benchmark eligibility or match individual bonds across sources.

between this corrected rate and observed issuance, when available. Fitting and selection between corrections use earlier years, separately from the year being evaluated.⁶ A second design removes the evaluated country’s entire history from fitting and selection to test whether a correction learned from other borrowers remains useful.

Table 10: Rating-implied accuracy with and without fitted corrections, 2018–2024

Estimation procedure	MAE	Reduction from original MAE
Original rating-implied formula	1.098	—
Correction using available country history	0.838	23.7%
Common correction with the evaluated country’s history withheld	1.005	8.5%

Notes: All rows evaluate the same 130 country-years in 40 countries during 2018–2024. MAE is in interest-rate percentage points relative to primary issuance. Unlike the policy comparison in Section 4.3, this exercise does not require a DAC reference. *Source:* Calculations from the constructed dataset.

The correction using available country history improves average accuracy in every evaluation year. With that history withheld, the common correction still lowers pooled MAE by 8.5%, but raises annual MAE in 2019 from 1.149 to 1.330 points and in 2021 from 0.801 to 0.990. These results support calibration among observed issuers, particularly when their earlier borrowing provides information for the correction. Withholding that information does not, however, establish performance for borrowers whose issuance rates remain unobserved.⁷

Of the 269 country-years in which a rating-implied rate is selected during 2018–2024, 241—about 90%—lack the earlier matched observations needed for a country adjustment. This thesis retains the original Treasury-plus-spread formula because the issuer-based corrections have not been validated for that wider population. This is a cautious choice about transferring a fitted correction, not a finding that the original formula minimizes pooled validation error. The calibrated estimates are reported as alternatives in this evaluation but are not used in the main analysis.

For peer estimates, this thesis examines missing rating information by recomputing the rates after withholding the target country’s recorded rating, using estimated creditworthiness and the matching rules in Chapter 3. Across 209 identical country-years in 46 countries during 2012–2024, MAE rises from 1.12 to 1.33 percentage points. Five otherwise evaluable country-years lose a peer estimate. The comparison shows weaker approximation when matching cannot use the recorded rating, a condition relevant to

⁶For each evaluation year, candidates are compared on earlier years using still earlier observations for fitting. The selected candidate is then refitted on all preceding observations. This separates model selection from evaluation, following the principle discussed by Cawley and Talbot (2010, p. 2080). The exercise remains retrospective, rather than a previously untouched confirmation sample. The country adjustment partially pools earlier country discrepancies, using variance components estimated with a random-intercept model in `nlme` (Pinheiro and Bates n.d.).

⁷Thirteen evaluation observations in thirteen countries have no earlier matched issuance and rating-implied observation in the dataset. In this small group, the common correction increases MAE from 0.94 to 1.19 points. This is limited accompanying evidence about borrowers lacking history, not a general finding for all such borrowers.

many borrowers needing peer estimates. However, it still contains only five low-income country-years, leaving limited direct evidence for that group.

4.6 Annual rate levels and changes over time

This thesis uses borrower-specific rates both to assess official-loan concessionality in a given year and to describe changes in borrowing conditions over time. To evaluate these uses separately, the analysis compares annual rates and their year-to-year changes with primary issuance for consecutive years within 2018–2024. Each comparison requires the primary rate, the alternative rate and the DAC reference in both years. All four alternatives have smaller average errors than DAC in annual rate levels on these observations, but only IDS also has a smaller average error in annual changes.

Table 11: Approximation of annual rates and their changes, 2018–2024

Alternative rate	Annual transitions (countries)	Level MAE: alternative	Level MAE: DAC	Change MAE: alternative	Change MAE: DAC
IDS bondholder terms	70 (22)	0.657	1.774	0.609	0.945
Secondary yields	66 (21)	1.185	1.733	1.463	0.957
Rating-implied rates	69 (22)	1.101	1.769	1.106	0.973
Peer estimates	75 (23)	1.157	1.743	1.142	0.988

Notes: MAE is in interest-rate percentage points relative to primary issuance. Both calculations use the same consecutive-year pairs within each row; samples differ between rows. Level MAE averages errors at both endpoints, so a year belonging to two transitions enters twice. Change MAE measures the error in the change between those endpoints. *Source:* Calculations from the constructed dataset using the error definitions in Section 4.1.

In 19 of the 75 peer comparisons, the peer rate is closer to primary issuance than the DAC rate in both years, yet its change is less accurate. The DAC reference is unchanged in all 19 cases, but that zero change is closer to the observed primary-rate movement. In 12 of these cases, the peer estimate switches from below the primary rate to above it, or vice versa, so its movement partly reflects a change in the direction of its error. The closer approximation of annual borrowing costs therefore coexists with a less accurate description of their change in these observations.

For analysis over time, this thesis consequently examines changes among the same countries and separately by source, alongside movements in the selected reference. These comparisons help identify the contribution of changing coverage and source availability, although they do not guarantee accurate changes. Each row above already holds the method fixed, and the bonds, observation dates or peer groups represented can still differ between years. The evidence also spans only six annual transitions, limiting conclusions about performance across other borrowing environments.

A separate check withholds primary issuance from selection and applies the IDS–secondary–rating–implied–peer order to the same issuer-year evidence. Across 75 transitions, the selected alternatives have level MAE of 0.688 points against 1.743 for DAC, and change MAE of 0.670 against 0.988. This

check mostly selects IDS and does not select peers, so it provides little evidence about the hierarchy’s performance for non-issuers dependent on estimated rates.

5 Borrowing conditions across countries and over time, 2012–2024

5.1 Changes in borrowing costs and their spread across countries

The borrowing-rate series for 2012–2024 shows a pronounced increase in 2022, accompanied by wider differences between lower- and higher-cost borrowers. This thesis calculates the annual average across low- and middle-income countries with an eligible primary, IDS, secondary or rating-implied reference, following the selection order in Chapter 3 and excluding countries supported only by peer estimates. Giving each country equal weight is a simplifying choice, with population or economic size providing alternative bases for weighting.

The mean rises from 5.31% in 2012 to 7.37% in 2018, with some intervening declines, before falling to 5.26% in 2020. It then increases in 2021 and sharply in 2022. Country coverage expands from 54 in 2012 to 85 in 2021 and falls to 71 in 2024, so comparisons retaining the same countries are used below to examine the main movements.

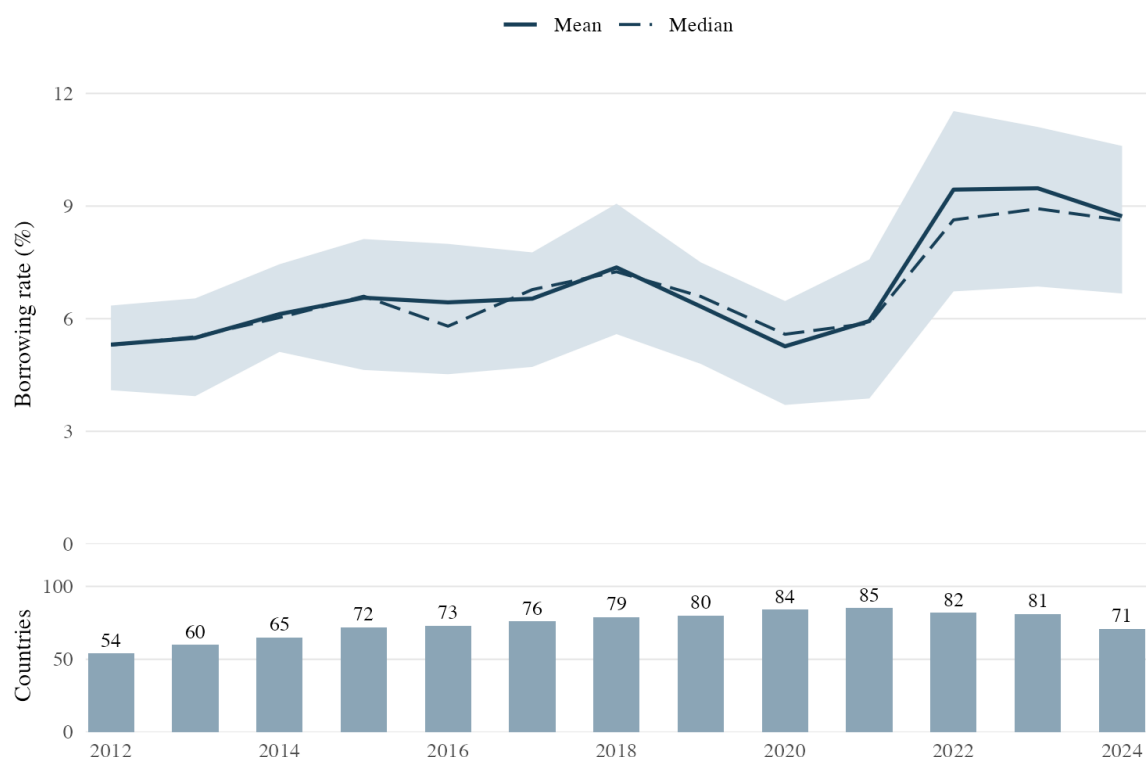


Figure 2: Borrowing rates and differences across countries, 2012–2024.

Notes: The lines show the equal-country mean and median of selected references, excluding peer-only cases. The shaded band spans the 25th to 75th percentiles and describes differences across borrowers in each year's sample. The lower panel reports country counts. *Source:* Calculations from the dataset constructed in Chapter 3.

Among the 81 countries represented in both 2021 and 2022, the mean rate increases by 3.44 percentage points, with increases in 77 countries. The rise also remains when the selected source is unchanged, and when primary, IDS, secondary and rating-implied rates are examined separately on their respective matched countries.⁸ Its timing coincides with US monetary tightening from 2022 (OECD 2025, 117). The OECD describes the accompanying rise in US yields, dollar appreciation and tighter financing conditions for emerging and developing economies (OECD 2025, 117–119).

Among those same 81 countries, the gap between the 25th and 75th percentiles widens from 3.55 percentage points in 2021 to 5.03 points in 2022. This widening also appears in each of the four separate rate sources on its own matched sample. The increase in average borrowing costs is therefore accompanied by greater differences between relatively lower- and higher-cost borrowers.

The annual mean is 9.47% in 2023 before falling to 8.73% in 2024. Among the same 70 countries, however, the 2024 decline is only 0.17 percentage points, indicating that changing country coverage contributes substantially to the larger fall in the annual average.

5.2 Regional differences and their evolution

The overall increase in borrowing rates is shared across regions, but their levels differ substantially. Sub-Saharan Africa and Latin America and the Caribbean have higher average rates than East Asia and the Pacific throughout 2014–2024. By 2024, the mean is 9.97% among the 23 represented sub-Saharan African countries, compared with 7.66% among the 11 East Asian and Pacific countries. The regional trajectories also show the uneven coverage behind these comparisons: the sub-Saharan African mean represents only six countries in 2012.

⁸Among the 49 countries retaining the same selected source in 2021 and 2022, the mean increase is 3.78 percentage points.

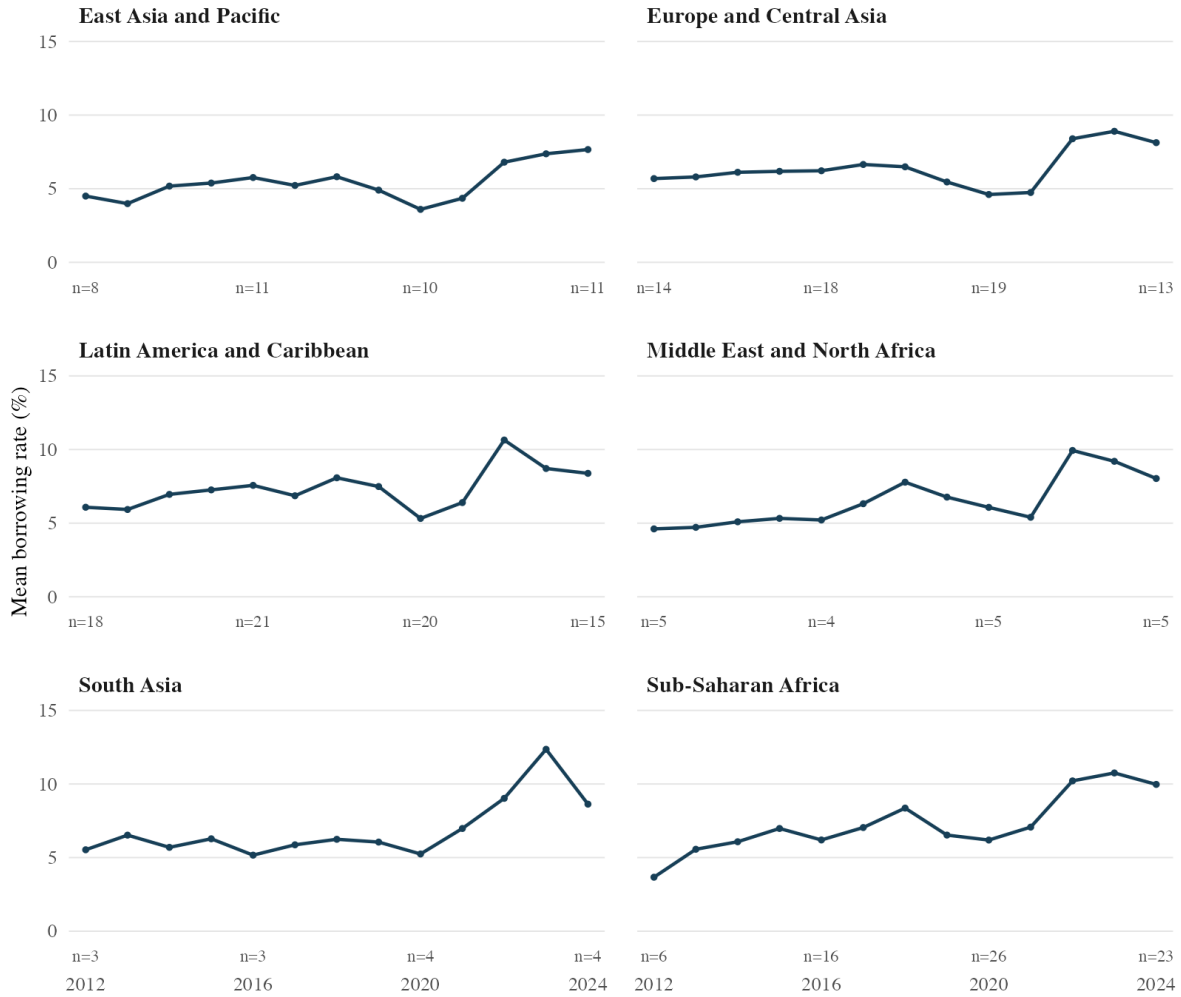


Figure 3: Regional borrowing rates, 2012–2024.

Notes: Equal-country means of selected references, excluding peer-only cases. All panels use the same percentage scale; country counts are shown for 2012, 2016, 2020 and 2024. Regional membership follows the World Bank grouping before July 2025 throughout, with Afghanistan and Pakistan in South Asia. *Source:* Calculations from the dataset constructed in Chapter 3.

OECD (2025, 121) likewise reports a rise in sub-Saharan Africa’s median secondary-market yield from around 6% in 2021 to nearly 10% in 2024. Its issuance comparisons weight bonds by issuance amounts, and its secondary comparisons use medians across countries; this thesis uses equal-country means and also includes IDS terms and rating-implied estimates.⁹

The 2021–2022 increase remains positive in every region when the same countries are compared. Its magnitude varies, with the smallest regional samples providing evidence for only four or five countries.

⁹The OECD secondary-yield figure carries 2023 values forward for 28 countries lacking a 2024 observation (OECD 2025, 121). Its reported 2024 regional medians therefore include earlier observations.

Table 12: Changes in borrowing rates within regions, 2021–2022

Region	Mean increase (percentage points)	Countries observed in both years
East Asia and Pacific	2.45	11
Europe and Central Asia	3.50	15
Latin America and Caribbean	4.25	20
Middle East and North Africa	4.54	5
South Asia	2.45	4
Sub-Saharan Africa	3.14	26

Notes: Equal-country means of changes in selected references, excluding peer-only cases. The selected source can differ between years. *Source:* Calculations from the dataset constructed in Chapter 3.

Using rating-implied rates in both years also produces a positive increase in every region. The broader comparison of 2020–2021 with 2022–2024 reaches the same conclusion when restricted to countries with a rate in every year: regional mean rates are higher in the later period across all six regions. Excluding East Asia and the Pacific raises the overall average modestly but leaves the sharp 2022 increase intact.

The combination of higher rates across regions and widening differences between borrowers shows why a single average provides an incomplete basis for assessing official lending terms. The financing conditions against which those terms are compared change both over time and across countries, whereas the DAC discount rate remains fixed within each recipient category. For an unchanged repayment schedule, the higher borrower-specific rates recorded after 2021 reduce the present value of repayments and increase the measured concession relative to market financing. Official credit consequently compares more favourably precisely when alternative borrowing becomes more expensive.

5.3 Borrower characteristics and differences in borrowing costs

Lower-middle-income countries have higher average borrowing rates than upper-middle-income countries in every year from 2012 to 2024 in the comparison excluding peer-only references. This thesis groups countries by their World Bank income classification in each year and retains equal-country weights. In 2024, mean rates are 11.25% for low-income countries, 9.33% for lower-middle-income countries and 7.73% for upper-middle-income countries, with respective samples of seven, 29 and 35 countries. Rates also differ within these groups. The middle half of lower-middle-income rates spans 8.24–10.75%, overlapping the upper-middle-income interval of 6.04–8.62%. Broad income categories therefore leave considerable variation in the borrowing rates against which official financing is assessed.

Table 13: Borrowing rates by income group, 2024

Income group	Countries	Mean rate (%)	Middle half of rates (%)
Low income	7	11.25	10.11–13.08
Lower middle income	29	9.33	8.24–10.75
Upper middle income	35	7.73	6.04–8.62

Notes: Selected primary, IDS, secondary or rating-implied references, excluding peer-only cases. The middle half runs from the 25th to the 75th percentile. Income groups follow the historical World Bank classifications used in Chapter 3. *Source:* Calculations from the dataset constructed in Chapter 3.

Six of the seven low-income rates in 2024 are rating-implied estimates, and the remaining rate is a secondary-market yield. Their mean consequently describes a small group represented mainly by estimated borrowing costs and cannot establish the average rate available to low-income countries generally. The difference between the two middle-income groups also appears in observed bond issuance: primary yields average 8.64% across nine lower-middle-income countries and 6.58% across fifteen upper-middle-income countries in 2024. The higher lower-middle-income average is thus present in actual borrowing as well as in the combined series, although these comparisons do not separate income from differences in creditworthiness, region or bond terms.

Gbohoui, Ouedraogo and Some (2023, 20) examine regulatory quality among the institutional characteristics associated with sovereign bond spreads. This thesis examines a related association using observed US-dollar issuance yields and the World Bank’s Regulatory Quality indicator, which measures perceptions of governments’ ability to formulate and implement sound policies and regulations (World Bank 2026). Across 215 country-years from 47 countries in 2012–2024, a one-unit higher indicator value is associated with a 1.46-percentage-point lower issuance yield after accounting for calendar-year and income-group differences.¹⁰ Better perceived regulatory quality is therefore associated with lower borrowing costs beyond the differences captured by broad income groups. This association does not establish how much a regulatory reform would reduce borrowing costs.

6 Official-loan data and concessionality calculations

This thesis examines how the concessionality of official loans changes when their repayments are discounted at borrower-specific rates instead of standardized policy rates. This chapter sets out the loan data, repayment assumptions and comparisons underlying the findings in Chapter 7, using the borrowing-rate evidence developed in Chapters 3–5.

¹⁰This thesis estimates an ordinary least-squares regression with calendar-year and income-group indicators, giving each country-year equal weight. It is a simplified descriptive specification motivated by Gbohoui and coauthors. The World Bank governance estimate uses an approximately -2.5 to $+2.5$ scale, with higher values indicating better perceived regulatory quality. The World Bank revised its historical regulatory-quality estimates in 2025. This analysis uses those revised values, which may differ from the estimates published when the bonds were issued.

6.1 Official-loan sources and the usable samples

This thesis uses loan records from the OECD Creditor Reporting System (CRS) to compare borrower-specific and standardized measures of concessionality across bilateral and multilateral creditors. CRS provides the broadest creditor and geographical coverage among the datasets examined here. Its reported interest rates, repayment dates, frequency and repayment type allow the calculation of the amounts due and when they fall due (OECD 2024; OECD n.d.-d). The sample includes new loans to recipient governments with positive commitments and sufficiently complete, consistent terms to reconstruct those payments.¹¹ Incomplete reporting limits coverage across providers and years. After matching the records to borrower-specific rates and excluding loans supported only by peer estimates, the sample contains 2,686 financing records across 85 countries and 22 providers. Of these, 1,274 fall in the central comparison period, 2018–2024.

Morris, Parks and Gardner (2020, 8–9) compare Chinese, IDA and IBRD lending by discounting repayments at a common 5% rate. This thesis recalculates concessionality using borrower-specific rates for usable records from 2012–2014, the years shared by their study and this thesis’s benchmark dataset. Later Chinese lending is examined using AidData’s records through 2023, restricted here to fixed-rate US-dollar loans to central governments or with sovereign guarantees (AidData 2025). The African Debt Database adds official lending to African central governments through 2024, covering Chinese and other bilateral and multilateral creditors (Manger et al. 2025, 11–12; African Debt Database 2026). Together, these records extend the comparisons to later borrowing conditions and different creditor mixes.¹²

The World Bank’s International Debt Statistics (IDS) also provide average interest rates, maturities and grace periods by borrowing country, year and creditor (World Bank n.d.-c). For China, IBRD and IDA, this thesis uses those averages to compare the concessionality of average lending terms, following ONE’s *Priced Out* approach (Harcourt, Haro Ruiz and Rivera 2026; ONE Data n.d.). Each observation describes a repayment schedule constructed from average terms, whose interpretation differs from an individual-loan result as explained in Section 6.2.

¹¹CRS sometimes divides a loan among activities (Manger et al. 2025, 11). This thesis combines World Bank entries with a shared loan identifier, year and compatible terms. Other records retain the CRS activity unit.

¹²The datasets partly share underlying loans, including ADD records drawn from CRS and AidData (Manger et al. 2025, 11–12). Results remain separate, and agreement between sources is interpreted with that overlap in mind.

Table 14: Financing records with usable terms and a borrowing benchmark other than peers

Data source	Years	Observation	Number	Countries
OECD CRS	2012–2024	Financing record	2,686	85
Morris, Parks and Gardner	2012–2014	Loan record	462	65
AidData	2015–2023	Loan record	164	37
African Debt Database	2015–2024	Loan record	1,624	29
IDS average terms: China, IBRD and IDA	2018–2024	Country-year-creditor scenario	594	81

Notes: The DAC-reference comparison includes 448 of the 462 Morris–Parks–Gardner records because it also requires recipient eligibility. The IDS row shows the central-period subset. Counts across sources and observation types should not be added together. *Source:* This thesis’s calculations from the sources described above and the borrowing benchmarks constructed in Chapter 3.

The borrower benchmarks primarily describe dollar borrowing, so loan currency affects the interpretation of the resulting concessionality estimates. CRS expresses commitment amounts in dollars for reporting, but the extract used here leaves contractual currency unidentified. The Morris–Parks–Gardner and ADD records include dollar loans alongside other or unidentified currencies. The calculations normalize each advance to 100 units, retain its repayment proportions and timing, and apply the primarily dollar-based borrowing reference to those payments. For non-dollar loans, this assumes a common currency without converting actual future payments at projected exchange rates. AidData’s sample consists of dollar-denominated loans, and a separate calculation restricted to 499 dollar-denominated ADD records provides a comparison within that currency. The CRS extract cannot support the same restriction.

6.2 Reconstructing repayment schedules

The concessionality comparison depends on both the amounts repaid and their timing, so this thesis reconstructs a payment schedule from the reported loan terms. Following the present-value approach described by Scott (2017, sections 1.2–1.4), each schedule is valued under the alternative discount rates. The calculation assumes the full loan is advanced at the outset and holds the recorded interest rate or service charge constant. For variable-rate loans, this is a simplifying assumption about future interest payments. In CRS, the reported repayment dates, frequency and repayment type determine whether principal is repaid in equal instalments or through a constant combined principal-and-interest payment (OECD n.d.-d). Interest is calculated on the outstanding balance and is also paid during grace periods. The advance is placed on the recorded commitment date. These schedules include principal and the interest or service charge represented in the rate input, excluding separately reported fees and actual staged disbursements.

For the Morris–Parks–Gardner, AidData and ADD records, this thesis models equal principal repayments twice yearly. The calculation builds on the grant-element approach used by Morris, Parks and Gardner (2020, replication code) and the repayment convention described by AidData (2025, 28–29). Using that calculation for ADD is this thesis’s adaptation. Some World Bank records in these sources label the period from first to final principal repayment as maturity. This thesis includes the preceding grace period

when reconstructing total loan duration, using archived dates or the source’s repayment-span definition. In the Morris–Parks–Gardner comparison, the effect of this reconstruction is assessed separately at their original 5% discount rate before changing to borrower-specific rates.

Following ONE’s method, the IDS comparison uses each country-year-creditor’s average terms to construct annual equal-principal repayments after grace, with interest on the remaining balance (ONE Data n.d., “The debt service schedule”). The resulting schedule describes a loan with those average terms. Its calculated grant element need not equal the average grant element of the actual loans because the present-value calculation is nonlinear (ONE Data n.d., note 8). The loan-record and average-term results therefore remain separate. Chapter 7 reports sensitivity to alternative interest-payment assumptions.

6.3 Discount rates and matching loans to borrowing benchmarks

To measure the effect of borrower-specific discounting, this thesis values each repayment schedule using both the eligible borrowing benchmark for the recipient’s commitment year and the standardized reference. The annual borrower rate applies throughout the schedule, with the loan terms and weights held identical across the two valuations. For bond-derived references, the quoted percentage is used numerically as the effective annual discount rate, without a compounding conversion. These are retrospective annual-reference comparisons: rates can include information recorded after commitment, and no loan-specific discount curve is fitted to the payment dates. The main loan comparisons exclude peer-only estimates. Additional results include those estimates or restrict the sample to primary issuance and IDS bondholder rates. Since these restrictions also change the loans represented, direct comparisons between rate methods use only loans for which both methods are available.

The main reference follows the DAC recipient categories applicable in the commitment year: 9% for least-developed and other low-income countries, 7% for lower-middle-income countries and 6% for upper-middle-income countries (OECD DAC 2014, Annex 2; OECD DAC n.d.). The central comparison covers 2018–2024, when grant-equivalent reporting became the headline ODA standard (OECD 2019). Applying the same category rule before 2018 permits a consistent comparison over 2012–2024, with those earlier values interpreted under the later framework. Separate calculations use the former DAC reference of 10% before 2018 and the common 5% rate used by Morris, Parks and Gardner (OECD DAC 2014, Annex 2, paragraph 3; Morris, Parks and Gardner 2020, 8–9). These references provide a common basis for assessing loan terms across creditors. The calculations measure concessionality rather than determining each loan’s contribution to official ODA statistics.

6.4 Summarizing the comparisons and assessing uncertainty

Chapter 7 reports the borrower-rate grant element minus the standardized-reference grant element, in percentage points. A positive difference means that the same financing appears more concessional relative to the borrower’s borrowing conditions. The present value of repayments per 100 advanced provides the equivalent presentation introduced in Chapter 2, following ONE (ONE Data n.d.). Equal-record averages describe the financing records represented in the sample, and commitment-weighted averages give larger

loans greater weight. Medians help assess the influence of extreme observations on the mean. The IDS comparison instead averages country-year-creditor scenarios. AidData also permits averaging across identified financing events, so that events with several records do not automatically receive more weight. Annual results accompany comparisons between 2018–2021 and 2022 onward. Since these periods contain different loans, comparisons retaining common countries, creditors or benchmark methods help assess the contribution of changing sample composition.

For AidData and ADD, approximate country-based 95% intervals assess the difference between the earlier and later period averages. The calculation repeatedly omits one country’s entire set of loans and recalculates that difference, allowing for related observations from the same borrower. Resampling whole countries provides an additional check (Cameron and Miller 2015, section II.F). These intervals hold the observed years, loan terms and borrowing-rate estimates fixed. The six or seven years available provide limited evidence about whether the period difference would recur under other borrowing conditions. CRS results are reported descriptively, with weighting and sample-composition checks.

7 How borrower-specific rates change measured concessionality

Using borrower-specific discount rates changes the measured concession in official lending, with substantial differences across loans and years. This chapter examines the size of those differences and how they relate to market borrowing conditions, official interest rates and repayment timing. Comparisons that hold repayment schedules fixed identify the effect of the discount rate. Comparisons of complete lending terms then examine how repayment timing affects the relative concessionality of different creditors.

7.1 Concessionality under borrower-specific and standardized rates

In the broad CRS sample, the reconstructed repayment schedules have lower mean grant elements under the annual borrower-specific references than under the differentiated DAC reference. Across 1,274 financing records from 2018–2024, the mean grant element is 46.09% under borrower-specific rates and 51.73% under the standardized reference, a difference of -5.64 percentage points. This main comparison excludes records supported only by peer estimates. Giving larger commitments greater weight preserves the direction and increases the difference to -6.41 points.

Table 15: Mean grant elements in the CRS sample, 2018–2024

Weighting	Borrower-rate grant element (%)	Standardized grant element (%)	Difference (percentage points)
Equal weight per financing record	46.09	51.73	-5.64
Weight by commitment amount	47.55	53.96	-6.41

Notes: Both rows contain the same 1,274 financing records. The difference is the borrower-rate grant element minus the standardized grant element. The standardized discount rate is 9%, 7% or 6%, according

to the recipient category described in Chapter 6. *Source:* This thesis’s calculations using OECD CRS records and the borrowing benchmarks constructed in Chapter 3 (OECD 2024; OECD n.d.-d).

Expressed as discounted repayments per 100 advanced, the equal-record averages are 53.91 under borrower-specific rates and 48.27 under standardized rates. This presentation follows ONE’s borrower-side repayment comparison (ONE Data n.d., “The Present Value Repayment Ratio”; Harcourt, Haro Ruiz and Rivera 2026). The average modeled grant element therefore remains substantially positive under the borrower-specific references. The standardized calculation assigns a larger concession to the same obligations on average.

The average conceals considerable variation in both directions. Borrower-specific rates produce a higher grant element for 467 records, or 36.7% of the sample. The median difference is -5.47 points, and the middle 80% of observations lies between -24.78 and $+12.63$ points. The mean absolute difference is 12.24 points, more than twice the magnitude of the signed mean. Opposing differences partly cancel when averaged, so the overall result understates how much the assessment changes for individual records. A uniform downward adjustment to standardized grant elements would consequently miss both the direction and magnitude of many borrower-specific differences.

The additional loan datasets also yield negative mean differences over the central period: -1.40 points across 68 AidData records from 2018–2023 and -1.08 across 1,176 African Debt Database records from 2018–2024 (AidData 2025; African Debt Database 2026). Their smaller averages concern different financing portfolios, with different recipients, creditors and repayment terms. Extending CRS to the full 2012–2024 period gives a mean difference of -7.37 points across 2,686 records. As established in Chapter 6, the pre-2018 values apply the later standardized category rule for analytical comparability.

7.2 Changes across years and loan samples

The annual CRS results show a marked narrowing and temporary reversal of the difference after 2021. Borrower-rate grant elements are approximately 15–16 points below standardized values in 2020–2021, then slightly above them in 2022–2023. The difference returns below zero in 2024, at -2.92 points. Averaging across the two periods gives -9.69 points for 2018–2021 and $+0.82$ for 2022–2024. The later-period mean therefore masks a change in direction within those three years, rather than establishing a persistent shift to higher borrower-rate concessionality.

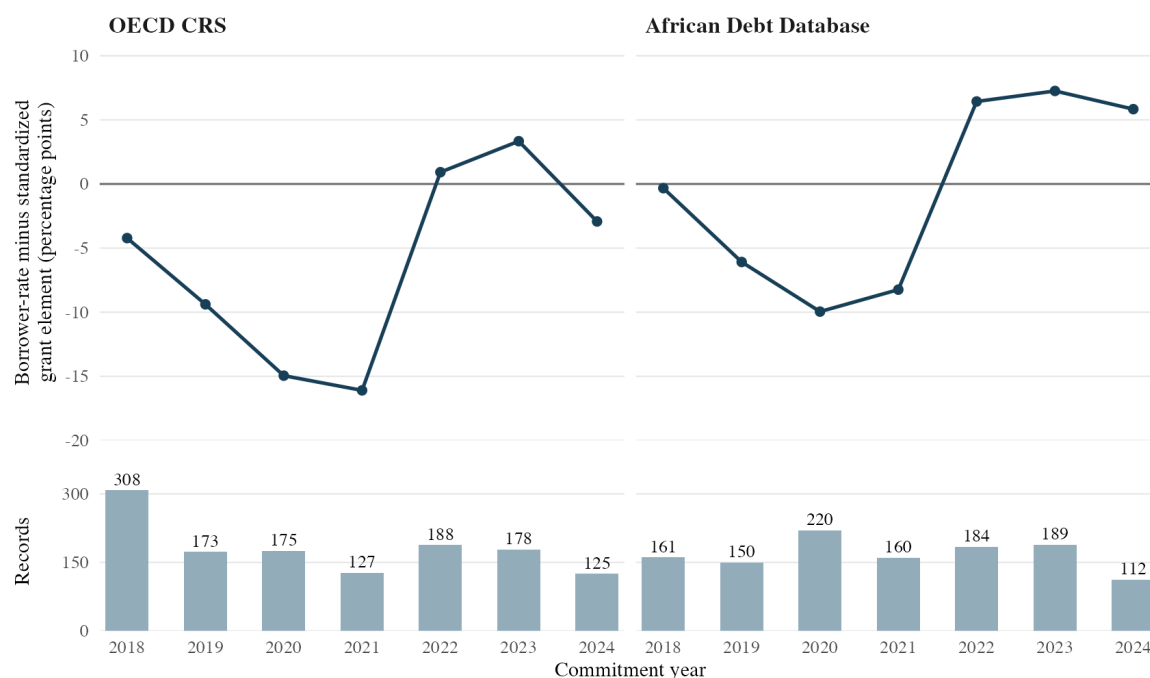


Figure 4: Annual differences between borrower-specific and standardized grant elements. Annual grant-element differences and financing-record counts for OECD CRS and the African Debt Database

Notes: Equal-record means, excluding records supported only by peer estimates. Positive values indicate higher concessionality under the borrower-specific rate. Counts identify the financing records represented in each year; borrowers and loans can differ between years. *Source:* This thesis's calculations using OECD CRS and African Debt Database records and the borrowing benchmarks in Chapter 3 (OECD 2024; African Debt Database 2026).

The small positive CRS average for 2022–2024 also depends on how financing is weighted. Giving larger commitments greater weight changes it from +0.82 to –0.93 points, indicating that the positive average across records does not extend to the average weighted by financing volume. Changing provider coverage does not fully explain the annual pattern: restricting the comparison to six providers represented in every year retains negative differences in 2020–2021, positive differences in 2022–2023 and a negative result in 2024. Recipients and loans still change within those providers.

The African Debt Database shows a stronger and more persistent later difference. Its mean rises from –6.48 points in 2018–2021 to +6.62 in 2022–2024, and remains positive in each of the three later years. The 13.09-point period increase is almost unchanged, at 13.12 points, when only countries represented in both periods are retained. Restricting the records to explicitly dollar-denominated loans also preserves a substantial increase, of 11.81 points. These checks show that neither changes in the set of countries nor inclusion of non-dollar loans is necessary for the positive period contrast. They do not, however, make the African and broader CRS portfolios directly comparable. The two datasets support different conclusions about the magnitude and persistence of the later gap.

The approximate country-based 95% interval for the African sample’s increase is 10.78–15.41 points, using the uncertainty calculation described in Section 6.4 (Cameron and Miller 2015, section II.F). It holds the observed years, rates and modeled terms fixed. AidData’s corresponding interval is much wider, from –10.40 to +21.48 points, and only six countries appear in both periods. Its later Chinese loan records remain useful for comparing concessionality, but provide weak evidence for a general temporal pattern.

7.3 Changing market borrowing conditions and official loan terms

The annual loan comparisons combine changes in borrowing conditions with changes in the official financing recorded. To examine those contributions separately, this thesis uses schedules constructed from IDS average terms and compares the same borrower and creditor in adjacent years. Each schedule is valued at both years’ borrower benchmarks. The contribution of the borrower rate is calculated once using the earlier repayment schedule and once using the later schedule, then averaged. The contribution of official terms is averaged in the corresponding way across the two discount rates. This follows the Shapley decomposition principle of averaging across update orders, which shares the interaction between the components (Shorrocks 1999, section 2). The resulting contributions explain changes in these modeled average-term comparisons, rather than decomposing the separate CRS loan sample.

Table 16: Contributions to changes in borrower-rate grant elements

Comparison	Matched borrower–creditor		Official repayment		Total change
	pairs	Borrower benchmark	terms		
2021–2022	65	+17.49	–3.00		+14.49
2022–2023	71	+1.27	–14.85		–13.58

Notes: Changes and contributions are in grant-element percentage points. Both years must have a benchmark other than a peer estimate. Each pair retains the same borrower and creditor, but describes new commitments in successive years. Contributions average both update orders and add to the total before rounding. *Source:* This thesis’s calculations using IDS average terms, ONE’s repayment construction and the borrower benchmarks in Chapter 3 (World Bank n.d.-c; ONE Data n.d.; Shorrocks 1999).

Between 2021 and 2022, the rise in borrower benchmarks contributes +17.49 points to the mean grant element, more than accounting for the total increase of 14.49 points. Changing official terms offsets part of that increase. Relative to the more expensive market alternative, official repayments become more favourable even though changes in their own terms reduce the concession. In 2022–2023, the benchmark contribution remains positive but is much smaller. Changes in official repayment terms instead account for a substantial decline in concessionality. Both interpretations persist when comparisons are restricted to the same benchmark method in adjacent years, although their magnitudes change. The components describe contributions to the calculated values, not identified causes of the underlying borrowing conditions.

The IBRD comparisons illustrate the importance of official interest rates. In a broader matched set of 36 country comparisons for 2022–2023, including peer-supported benchmarks, average recorded interest on

new IBRD commitments rises from 1.65% to 5.71%. Separating official interest from maturity and grace attributes -30.47 grant-element points to the interest change. Borrower benchmarks and maturity/grace contribute $+0.16$ and $+1.33$ points, leaving a total decline of 28.98 points.¹³ These figures describe new commitments in successive years, rather than interest resets on the same outstanding loans. Holding each recorded rate constant throughout the modeled loan also makes the estimated concession sensitive to future interest assumptions. A one-percentage-point change in interest after the first year changes mean IBRD grant elements by roughly 6.5–8.6 points in the sensitivity calculations. The average absolute change in the borrower-minus-standardized difference is smaller, at about 1.2–1.4 points, because both valuations receive the same altered repayments.¹⁴

The distinction between the concession itself and its difference from the standardized calculation changes the interpretation of the 2023 result. In the matched comparison excluding peers, the borrower-rate grant element falls by 13.58 points, yet the borrower-minus-standardized difference rises by 3.13 points. Both assessments show declining concessionality, but the standardized grant element falls further. The increasing difference between the measures therefore does not indicate an improvement in the official financing terms.

7.4 How the borrowing-rate estimate affects measured concessionality

Alternative estimates of the same borrower’s financing conditions can materially change the assessed concession even when repayment terms are identical. Across 92 country-year-creditor comparisons in 2022–2024 with both rating-implied and primary or IDS benchmarks, using the rating-implied rate raises the mean grant element by 7.25 points. The mean absolute change is 9.40 points. These differences isolate the consequence of benchmark choice because the observations and repayment schedules remain fixed. Chapter 4’s comparison against observed issuance supplies the separate evidence for judging accuracy. A larger calculated concession is not itself evidence that a benchmark is more accurate.

The African Debt Database’s positive period contrast also persists when a single benchmark method is used throughout. Among records with each method available, the increase in the borrower-minus-standardized difference between 2018–2021 and 2022–2024 is 8.30 points with IDS rates, 11.30 with secondary yields and 14.67 with rating-implied rates. The respective samples contain 429, 417 and 1,083 records. Changing the selected method is therefore not necessary for the positive contrast, although its size remains sensitive to the method and the loans it covers. Because these samples differ, the three magnitudes do not establish an ordering of benchmark performance.

Broadening or narrowing the borrowing-rate evidence also changes which loans can be assessed. In CRS, including records supported only by peers leaves the mean difference close to the main result. Restricting

¹³With three components, the calculation averages their contributions across all six update orders, applying the same Shapley principle (Shorrocks 1999, section 2).

¹⁴The sensitivity raises or lowers interest by one percentage point after year one, with a zero floor, across 262 IBRD scenarios. The ranges summarize the two directions separately for 2018–2021 and 2022–2024.

the sample to primary and IDS benchmarks produces a considerably more negative mean and removes almost two-fifths of the main sample.

Table 17: CRS comparisons under different benchmark-availability restrictions, 2018–2024

Borrowing benchmarks admitted	Financing records	Mean grant-element difference (percentage points)
Primary, IDS, secondary and rating-implied	1,274	−5.64
The same sources, plus peer estimates	1,531	−6.02
Primary and IDS only	798	−10.75

Notes: Equal-record averages of borrower-rate minus standardized grant elements. The samples differ between rows. *Source:* This thesis’s calculations using OECD CRS records and the borrowing benchmarks in Chapter 3 (OECD 2024; OECD n.d.-d).

The negative CRS average is therefore present across all three samples, but its magnitude depends on the coverage admitted. Unlike the identical-schedule comparison above, these differences combine the available borrowing-rate evidence with changes in the financing records represented. The primary-and-IDS result consequently describes the subset with those borrowing observations, rather than an alternative estimate for the entire main portfolio.

7.5 Repayment schedules and comparisons between creditors

A lower contractual interest rate can be offset by earlier principal repayments, changing which creditor’s terms provide the larger concession. Among 177 borrower-year creditor pairs with benchmarks other than peers, the ordering based on interest rates reverses under complete repayment valuation in 17 cases. The median absolute grant-element difference among those reversals is 5.68 points. Including peer-supported comparisons gives 19 reversals among 209 pairs. These results use schedules constructed from the creditor-specific average terms in IDS, following ONE’s approach (World Bank n.d.-c; ONE Data n.d., “The debt service schedule”).

Côte d’Ivoire in 2024 illustrates the mechanism. The recorded average interest rate is lower for Chinese lending than for IDA, at 2.00% against 2.25%. Chinese maturity is approximately 19.9 years, compared with 27.5 years for IDA, and the average grace period is also shorter. Valuing both schedules with the same primary borrowing benchmark gives grant elements of 47.49% for China and 53.17% for IDA. The lower Chinese interest rate does not compensate for the earlier repayments. Interest-rate comparisons alone would therefore give the opposite ordering for these average terms.

Across common borrowers and years in 2018–2024, the Chinese average-term scenarios have lower mean modeled grant elements than the IBRD and IDA scenarios by 13.88 and 24.29 points, respectively, when peer benchmarks are excluded. These comparisons contain 62 and 67 borrower-year pairs. IBRD’s mean grant element is 8.66 points below IDA’s across 48 matched pairs, although IBRD has the higher value in 21 of them. The averages thus describe tendencies within the recorded portfolios, with considerable

variation among borrowers. Matching the borrower and year gives both creditors the same discount rate, but does not match their projects or lending purposes.

The original records used by Morris, Parks and Gardner allow a separate examination of how borrower-specific discounting changes an established creditor comparison. Their study uses a fixed 5% rate and explicitly discusses the scope for borrower-specific rates (Morris, Parks and Gardner 2020, 8–9). Across 462 usable records from the overlapping 2012–2014 period, excluding peer-only benchmarks, replacing 5% with borrower-specific rates raises the mean grant element by 2.95 points. Both calculations use the same reconstructed repayment schedules, so that difference is separate from the effect of reconstructing loan duration described in Chapter 6.

For borrowers receiving financing from both creditors in the same year, the China-minus-IBRD difference changes from -18.85 points at 5% to -17.98 under borrower-specific rates across eight borrower-years. The China-minus-IDA difference changes very little across nine borrower-years. Within this small matched sample, incorporating borrower conditions changes the measured concession without overturning the average Chinese–World Bank ordering. The extension therefore adds information about how favourable the terms are relative to recipients’ financing conditions, while retaining the direction of the creditor comparison on the observations that can be matched.

8 Discussion and conclusion

8.1 Concessionality, additionality and development effectiveness

Research on development effectiveness asks what financing achieves beyond what would have happened without the intervention. In the additionality literature, Carter, Van de Sijpe and Cael (2021, section 2.2) examine whether development finance institutions enable investment that would otherwise not occur. Their analysis concerns investment in private-sector projects, but the underlying question also motivates scrutiny of sovereign development lending: what does official financing enable the recipient to do that its other financing options would not? Assessing the financial advantage offered to the borrower is one part of investigating that question.

Mutambatsere and Schellekens (2020, 6–8, 15, 29) connect this question explicitly to concessionality. In their analysis of blended finance, they distinguish the development value of a project, the additional contribution of a development finance institution and the need for concessional terms. A project may produce worthwhile outcomes without requiring subsidized financing, and an institution may add value through expertise or access to finance without offering a financial concession. The effectiveness question therefore concerns what the concessional component itself enables and how much is needed. Although their framework addresses private investment, it provides a useful basis for asking how favourable sovereign lending terms contribute to development outcomes.

This thesis contributes to that research by comparing the present value of modeled official repayments under borrower-specific annual references across a broad country-year panel. The results show why nominal loan amounts or standardized grant elements alone can give an incomplete account of the

concession received. The resulting differences show how much the assessed concession can depend on the borrowing reference and repayment assumptions. The panel provides a starting point for those comparisons under substantial constraints on data availability and comparability.

Linking this panel to development outcomes would allow further research to ask whether more concessional financing produces larger additional gains, and how those gains compare per unit of concession. Establishing those effects would require outcome data and a design that separates the consequences of the financing terms from the conditions that led to those terms. The financial denominator would also need to match the research question: a borrower-side concession measures the recipient's financing advantage, whereas effectiveness per unit of public expenditure requires the donor's cost (Potts and Chung 2008, 1027–1029).

8.2 Public communication and aid accounting

Expressing the present value of repayments per 100 advanced gives journalists, civil-society organisations and other non-specialist audiences an accessible way to compare sovereign lending terms. This thesis follows ONE's presentation of the borrower-side comparison (ONE Data n.d.; Harcourt, Haro Ruiz and Rivera 2026). Stated alongside the loan amount and borrowing benchmark, the repayment value makes the degree of concessionality more concrete than a financing-volume announcement alone. This can give public discussion of lender–borrower relations a clearer financial basis, with the underlying terms and reference rate available for scrutiny.

If accounting rules assigned lower grant equivalents to eligible loans, recorded ODA would fall for unchanged disbursements and increase the shortfall of a donor already below the 0.7% of gross national income target (OECD n.d.-c; OECD n.d.-e). The main CRS comparison produces lower average grant elements when borrower-specific rates replace the differentiated policy rates, without recalculating individual donors' ODA totals.

8.3 Limits of interpretation and priorities for further research

The accuracy evidence is strongest for countries that issue bonds, because their observed issuance rates provide the basis for evaluation. The dataset also contains estimates for countries without those observations, including borrowers whose financing alternatives may be more restricted. Performance among observed issuers therefore provides incomplete evidence about accuracy for the countries most dependent on estimates. Peer medians can themselves rely heavily on rating-implied inputs and combine rates with different timing and currency coverage; their additional coverage is not independent market evidence. Validation on later borrowing observations for previously unobserved countries would strengthen the basis for using these estimates. This thesis holds observed or estimated borrowing conditions fixed when evaluating official repayments. Assessing whether a government could replace a substantial volume of official finance with market borrowing would require evidence on financing availability, instrument terms and how borrowing costs respond to additional debt.

Closer matching between each loan and its borrowing benchmark would improve the comparison of financing terms. The annual rates used here do not reproduce every loan’s currency, maturity and commitment date, and the broad CRS records do not establish contractual currency. The 5.64-point mean difference is consequently a result under the common-currency, annual-rate and repayment assumptions, not an estimate of the gap from fully matched market financing for each loan. Chang, Fernández-Arias and Servén (1999, 11–13) used currency- and maturity-specific government-security yield curves to measure donors’ financial concessions. An extension could combine that detail with borrower-specific risk premia and information available at commitment. Each payment would then be valued using a discount factor matched to its currency and horizon. More complete contractual data would also permit actual disbursement schedules, fees and variable interest payments to replace the present assumptions. The interest-rate sensitivities in Chapter 7 show that repayment assumptions can materially affect grant-element levels, although their effect on the difference between two valuations of the same schedule is smaller in that exercise.

Two implementation choices also require further assessment: the rating estimate maps foreign-currency grades to spread groups based on local-currency ratings, and the repayment calculations use quoted bond yields without harmonizing their compounding conventions. Further research should align the rating concepts and convert yields to a common effective annual basis before aggregation and discounting. The secondary series would also benefit from dated historical outstanding balances and a common date for measuring remaining maturity across its two calculation routes. The effect of these changes on the headline concessionality difference has not been quantified.

The results also depend on which financing records contain usable terms and can be matched to a borrower benchmark. They describe those portfolios, with common-country and common-provider comparisons testing specific changes in composition. The analysis does not reweight the sample to represent excluded CRS lending. Comparing included and excluded records by provider, recipient risk and loan terms would help establish the direction of that selection. The reported country-based uncertainty intervals account for dependence among observations from the same country, following the reasoning discussed by Cameron and Miller (2015), but hold the benchmark estimates and modeled repayments fixed. They consequently leave uncertainty about those inputs outside the intervals. Shared rate inputs, the short span of calendar years and loans appearing in more than one dataset also limit the independent evidence available. A further extension would propagate alternative benchmark and repayment assumptions jointly through the calculations. The present separate sensitivities do not establish a probability that the headline difference or each creditor ordering retains its sign under that combined uncertainty.

8.4 Conclusion

This thesis constructs borrower-specific sovereign borrowing benchmarks for 2012–2024 by combining observed financing terms and market yields with estimates for countries lacking direct observations. Retaining the sources and assumptions makes it possible to examine both coverage and comparability. In the matched 2018–2024 issuance samples, all four alternatives to primary yields have a lower mean

absolute error than the standardized policy reference, although the strength of the evidence varies across methods. The evaluation provides stronger support for comparing annual borrowing-rate levels than for measuring changes from one year to the next. These results establish both the usefulness of borrower-specific information and the limits of treating the resulting rates as interchangeable observations.

Using those annual benchmarks changes the concessionality calculated from the reconstructed official repayment schedules. In the main CRS comparison for 2018–2024, the mean grant element is 5.64 percentage points lower than under the differentiated policy rates. Substantial differences across financing records and years mean that this average cannot serve as a uniform adjustment. The findings also show how changes in official lending terms and the timing of repayments affect the financing advantage and comparisons between creditors. By constructing and evaluating the borrowing benchmarks, then applying them to recorded official lending terms, this thesis makes recipients’ financing conditions explicit in concessionality measurement. The resulting evidence provides a basis for research on the effectiveness of concessional finance and for clearer public comparisons of sovereign lending terms.

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